

THE TOPOLOGY IS THE SINK: TOPOLOGICAL HAL-LUCINATION DETECTORS REDUCE TO FIRST-ORDER ATTENTION STATISTICS

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ABSTRACT

Persistent homology (PH) of attention graphs underlies a growing family of hallucination detectors, including the recent TOHA (ACL 2026). We show that, in decoder-only models, these topological features reduce to first-order attention statistics dominated by the *attention sink*, either exactly or up to one computable number. For any symmetric dissimilarity and vertex s , we define the *coning defect* δ_s . Combining classical cone and stability arguments, we show that the entire Vietoris–Rips barcode lies within bottleneck distance δ_s of a “star diagram” read off the distances to s . Every stable vectorization is therefore a function of one column of the attention matrix, up to δ_s . Applied to TOHA, the same argument shows that its prompt–response topological divergence *equals* the mean, over response tokens, of one minus their largest prompt attention whenever a prompt-level coning defect vanishes, and lies within that defect otherwise. On 18,906,944 per-head graphs from 16 settings (7 open models of 1.1B–7B parameters; TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark we build), this identity has no violations and is exact on 57–88% of them. Adding TOHA’s score to its first-order counterpart changes AUC by at most 0.007 (equivalent within ± 0.015 in 15 of 15 settings), and supervised probes on the two are equivalent in 15 of 15. For whole-graph PH, 838,356 layer graphs show no bound violations; in every model with a first-token sink, 36–98% are *exactly* coned, 1D persistence is zero on all of them, and length-normalized 0D persistence keeps a median per-layer rank correlation of at least 0.98 with sink attention. For detection, on TruthfulQA and on-policy TriviaQA, none of 0D PH, sink-deflated PH, or the per-head coning defect adds more than 0.009 AUC to cheap non-topological probes, and TOHA adds at most 0.013. We release the extractor, evaluator, and all per-example features.

1 INTRODUCTION

A growing family of hallucination detectors treats a transformer’s attention matrix as a weighted graph and summarizes it with persistent homology (PH): barcodes of connected components (H_0) and loops (H_1) along a Vietoris–Rips filtration (Kushnareva et al., 2021; Cherniavskii et al., 2022; Perez & Reinauer, 2022; Kostenok et al., 2023; Bazarova et al., 2026; Samaga et al., 2026). The appeal is that topology might capture *global* structure in how information flows that token-level statistics miss. A separate line of work has established that decoder-only language models route a large share of attention to the first token, the *attention sink* (Xiao et al., 2024; Sun et al., 2024; Cancedda, 2024; Gu et al., 2025; Barbero et al., 2025). Only recently have the two been connected: Binkowski et al. (2026) show that sinks carry much of the signal used by attention-based hallucination detectors. No work has yet asked what sinks do to the *topology* of attention graphs.

We show that, for topological detectors, the two literatures study the same object to a large and quantifiable extent. When one vertex s is closer to every other vertex than those vertices are to each other, every sublevel set of the Vietoris–Rips filtration is a cone with apex s : all higher homology vanishes and the minimum spanning tree (which computes H_0) is the star around s . Real attention graphs sit at or very near this configuration, because of the sink. The “topology” of a causal attention graph is therefore, in most layers, exactly a function of how much attention each token sends to the sink, and in the remaining layers is controlled by a single computable number, the coning defect.

The same holds for the published detector that restricts topology to the prompt–response interface: TOHA’s divergence is a first-order attention statistic plus a computable correction, and on real heads the correction adds at most 0.007 AUC.

Contributions.

1. **A whole-barcode sink reduction** (§3). The ingredients are classical: a flag complex with a dominating vertex is a cone (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and persistence is stable (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). We combine them into one-sided bounds in a single computable number. For any symmetric dissimilarity and vertex s , bars of dimension $k \geq 1$ have length at most δ_s , and 0D total persistence is sandwiched around the star weight (Theorem 1). The whole barcode is within bottleneck distance δ_s of the star diagram, so every stable vectorization reduces to one column of distances (Corollary 1). For causal attention, δ_0 has a closed form (Corollary 2).
2. **The published topological detector is a first-order detector** (§3, §5.4). TOHA’s topological divergence equals one minus the response tokens’ mean max-prompt attention on prompt-coned heads, and is within a computable defect of it otherwise (Proposition 1). This makes exact the qualitative remark of Binkowski et al. (2026). On 18,906,944 real head graphs the identity has no violations, and adding TOHA to its first-order counterpart changes AUC by 0.000 to +0.007.
3. **Verification of the whole-graph reduction** (§5.1, Appendix D). There are zero bound violations in 838,356 graphs. Graphs are exactly coned in 36–98% of cases in sink-dominated models, H_1 is empty on every coned graph, and a sink dose-response experiment traces the weakness of 1D features to coning.
4. **A detection audit with strong baselines** (§5.3). On TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark, we compare 0D/1D PH, sink features, sink-deflated PH, the coning defect, and TOHA against length, lexical, log-probability, LLM-Check (Sriramanan et al., 2024), Lookback-Lens (Chuang et al., 2024), and hidden-state probes, with grouped cross-validation, early- and late-fusion incremental tests, and equivalence tests.

2 BACKGROUND AND RELATED WORK

Topology of attention. Kushnareva et al. (2021) and Cherniavskii et al. (2022) introduced barcodes of symmetrized BERT attention graphs as features for artificial-text detection and acceptability; Perez & Reinauer (2022) and Kostenok et al. (2023) extended them to classification and uncertainty estimation. For hallucination detection, TOHA (Bazarova et al., 2026) computes an MST-based divergence between prompt and response subgraphs, HalluZig (Samaga et al., 2026) uses zigzag persistence across layers, and CHARM (Frasca et al., 2026) learns a message-passing model over attention graphs. Gardinazzi et al. (2025) study zigzag persistence of hidden representations. All 0D summaries in this family are MST statistics (Fact 1; Gower & Ross, 1969); our results concern what those statistics measure in *causal* models.

Attention sinks. Xiao et al. (2024) observed that streaming LMs keep the first tokens’ key-values because a large share of attention lands there; Sun et al. (2024) tied sinks to massive activations, Cancedda (2024) to low-rank “dark signals”, Gu et al. (2025) characterized when they emerge during pre-training, and Barbero et al. (2025) argued they prevent over-mixing; related “no-op” attention to special tokens was observed in BERT (Clark et al., 2019) and vision transformers (Darcet et al., 2024). Binkowski et al. (2026) use sinks with large-norm values directly as a hallucination signal (SinkProbe) and argue that earlier attention-based detectors rely on them implicitly. Our theorem makes this precise for topological detectors, turning sink strength into a statement about the topology of attention graphs.

Hallucination detection. Beyond topology, detectors use sampling consistency (Manakul et al., 2023; Farquhar et al., 2024), internal probes (Azaria & Mitchell, 2023; Burns et al., 2022; Marks & Tegmark, 2023; Li et al., 2023b; Kossen et al., 2024; Orgad et al., 2024; Du et al., 2024), covariance spectra (Chen et al., 2024), attention-kernel log-determinants (Sriramanan et al., 2024), and context-vs-generated attention ratios (Chuang et al., 2024). Recent audits find that several benchmarks admit near-perfect detection from surface form alone (Hussain & Kantarcioglu, 2026; Janiak et al., 2025),

echoing annotation artifacts in NLI (Gururangan et al., 2018; McCoy et al., 2019); we therefore include lexical controls and an on-policy benchmark.

3 THEORY: ATTENTION SINKS CONE THE VIETORIS–RIPS FILTRATION

Let $D \in \mathbb{R}_{\geq 0}^{N \times N}$ be symmetric with zero diagonal. The Vietoris–Rips filtration is $K_\epsilon = \{\sigma \subseteq [N] : \max_{u,v \in \sigma} D_{uv} \leq \epsilon\}$; $\text{dgm}_k(D)$ is the barcode of $H_k(K_\epsilon; \mathbb{F})$ over a field, and $P_k(D)$ is its total persistence (sum of finite bar lengths). For attention, $A^{(l)}$ is the head-averaged, row-stochastic, lower-triangular attention matrix of layer l and, following the literature, $D = 1 - \max(A, A^\top)$ with zero diagonal.

Fact 1 (Gower & Ross, 1969). *All 0D bars are born at 0, and their finite death times are, as a multiset, the edge weights of a minimum spanning tree (MST) of D . In particular $P_0(D) = w(\text{MST}(D))$.*

Definition 1 (Coning defect, star weight). For $s \in [N]$, the *coning defect* is $\delta_s(D) = \max_{u \neq v, u, v \neq s} [\max(D_{su}, D_{sv}) - D_{uv}]_+$ (zero if $N < 3$), and the *star weight* is $S_s(D) = \sum_{u \neq s} D_{su}$. We call D *coned at s* if $\delta_s(D) = 0$, i.e. s is at least as close to both endpoints of every edge as they are to each other.

Theorem 1 (Sink-coning reduction). *For every symmetric D and every vertex s , with $\delta = \delta_s(D)$:*

- (a) *every bar of $\text{dgm}_k(D)$, $k \geq 1$, has length at most δ ; hence $P_k(D) \leq \delta \cdot |\text{dgm}_k(D)|$;*
- (b) *$S_s(D) - (N-1)\delta \leq P_0(D) \leq S_s(D)$, and $\max_u D_{su} - \delta \leq \max \text{dgm}_0^{\text{fin}}(D) \leq \max_u D_{su}$;*
- (c) *if D is coned at s , then $H_k(K_\epsilon) = 0$ for all $k \geq 1$ and all ϵ , the star at s is an MST, and $P_0(D) = S_s(D)$.*

Proof sketch (full proof in Appendix A). Let $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$ and $D'_{su} = D_{su}$. Then $D \leq D' \leq D + \delta$ entrywise and D' is coned at s . Since D' is coned, any simplex $\sigma \not\ni s$ with $|\sigma| \geq 2$ has the same filtration value as $\sigma \cup \{s\}$, and a vertex not yet joined to s has no edges; so each K'_ϵ is a cone with apex s plus isolated points, and $H_k(K'_\epsilon) = 0$ for $k \geq 1$. The inclusions $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$ factor the persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ through zero, so no bar outlives δ . For (b), the star is a spanning tree of D (upper bound) and an MST of D' ; MST weight and bottleneck are $(N-1)$ - and 1-Lipschitz in the sup-norm. The ingredients are standard: Lemma 1 is the dominating-vertex (cone) argument behind strong and edge collapses (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and the last step is a one-sided form of the interleaving argument behind persistence stability (Chazal et al., 2009). $\square$

The bound in (a) is attained: if $D_{su} = 1$ for all u and four further vertices form a 4-cycle with side $\frac{1}{2}$ and diagonals 1, then $\delta_s = \frac{1}{2}$ and H_1 has the single bar $[\frac{1}{2}, 1)$.

Theorem 1 controls summaries; the same comparison controls the whole barcode. Call $\Sigma_s(D)$ the *star diagram*: in degree 0 the bars $[0, D_{su})$, $u \neq s$, plus one infinite bar, and nothing in degree $k \geq 1$.

Corollary 1 (The whole barcode is the star, up to δ_s). *For every k , $d_B(\text{dgm}_k(D), \Sigma_s(D)_k) \leq \delta_s$, with equality of the diagrams when $\delta_s = 0$. More precisely, the j -th smallest finite 0D death of D lies in $[D_{s(j)} - \delta_s, D_{s(j)}]$, where $D_{s(j)}$ is the j -th smallest entry of $(D_{su})_{u \neq s}$, and β_0 of K_ϵ lies between $1 + \#\{u : D_{su} > \epsilon + \delta_s\}$ and $1 + \#\{u : D_{su} > \epsilon\}$. Hence every vectorization that is L -Lipschitz in d_B (e.g. persistence landscapes; Bubenik, 2015) is within $L\delta_s$ of a function of the column $(D_{su})_u$. Vectorizations that are stable in 1-Wasserstein distance (e.g. persistence images; Adams et al., 2017) are within $L((N-1)\delta_s + \frac{1}{2}\delta_s \sum_{k \geq 1} |\text{dgm}_k(D)|)$ of such a function. On coned graphs, every vectorization, including persistence entropy and Betti curves, is exactly such a function.*

Corollary 2 (Causal attention). *Let A be lower-triangular and row-stochastic, $D = 1 - \max(A, A^\top)$, and $s = 0$ the first token. Then $D_{0u} = 1 - A_{u0}$, so $S_0 = (N-1)(1 - \bar{m})$ with $\bar{m} = \frac{1}{N-1} \sum_{u \geq 1} A_{u0}$ the mean sink attention, and*

$$\delta_0 = \max_{1 \leq v < u} [A_{uv} - \min(A_{u0}, A_{v0})]_+.$$

Thus $P_0 \in [(N-1)(1 - \bar{m} - \delta_0), (N-1)(1 - \bar{m})]$ and all $H_{k \geq 1}$ bars have length at most δ_0 . The graph is exactly coned iff, for all $1 \leq v < u$, $A_{uv} \leq A_{u0}$ and $A_{uv} \leq A_{v0}$: no token attends to an earlier non-sink token more than either of the two attends to the sink.

Remark 1 (Choice of distance). Coning is an order condition, so it is invariant under any entrywise strictly increasing transform f of D with $f(0) = 0$ (e.g. $-\log$ of the symmetrized attention, or ranks without ties, in place of $1 - A$). The set of exactly coned graphs, and the vanishing of their higher homology, does not depend on how attention is turned into a distance; only the magnitude of δ_s on non-coned graphs does. (A merely non-decreasing f , such as clipping, can create coning.)

Corollary 2 makes four testable predictions: **(C1)** in coned layers 0D total persistence equals $(N-1)(1 - \bar{m})$, so after dividing by $N-1$ it is an affine function of sink attention; **(C2)** H_1 is empty whenever $\delta_0 = 0$; **(C3)** 1D persistence can only be non-zero where the sink is violated, with bars no longer than δ_0 ; **(C4)** removing the sink and renormalizing (*deflation*) exposes whatever non-sink topology exists.

Random weights and directed complexes. For i.i.d. weights, by contrast to Theorem 1, 1D total persistence grows linearly in N (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); we give an elementary explicit bound, $\mathbb{E}[P_1] \geq \frac{3}{8}N - \sqrt{3N} + O(1)$ (Proposition 2, Appendix A). For causal attention the directed flag complex is no escape: the digraph is a transitive tournament, so its weighted directed flag complex is isomorphic to $\text{VR}(D)$ as a filtered complex (Proposition 3, Appendix A.4).

The published topological detector reduces to a first-order statistic. TOHA (Bazarova et al., 2026) scores an attention head by the *topological divergence* between the response tokens R and the prompt tokens P . Its per-head score is $d(R, P) = \text{MTopDiv}(R, P)/|R|$, where MTopDiv is the total 0D persistence of the head’s graph $D = 1 - \max(A, A^\top)$ after all prompt–prompt distances are set to 0 (equivalently, the minimum spanning forest attaching R to P). TOHA averages d over at most ten heads chosen on labelled data. For $u \in R$ let $\pi_u = \max_{v \in P} A_{uv}$ be u ’s largest attention to any prompt token.

Proposition 1 (TOHA is max-prompt attention, up to a defect). *Let A be lower-triangular and row-stochastic, and let $P = \{0, \dots, p-1\}$ and $R = \{p, \dots, N-1\}$. With $\delta_P = \max_{w < u \in R} [A_{uw} - \min(\pi_u, \pi_w)]_+$,*

$$\frac{1}{|R|} \sum_{u \in R} (1 - \pi_u) - \delta_P \leq d(R, P) \leq \frac{1}{|R|} \sum_{u \in R} (1 - \pi_u),$$

with equality when $\delta_P = 0$. Moreover $1 - \pi_u \leq 1 - A_{u0}$, with equality iff the first token is among u ’s most-attended prompt tokens. On a head where this holds for every $u \in R$ and $\delta_P = 0$, $d(R, P) = 1 - \frac{1}{|R|} \sum_{u \in R} A_{u0}$, the response tokens’ mean sink attention.

Proof sketch. Zeroing the prompt block merges P at scale 0, so the remaining finite 0D bars are the MST weights of the quotient graph on $\{p^*\} \cup R$, where $D_{p^*u} = \min_{v \in P} D_{uv} = 1 - \pi_u$ (Fact 1). Apply Theorem 1(b) at the apex p^* ; as in Corollary 2, its coning defect is δ_P and its star weight is $\sum_u (1 - \pi_u)$. Divide by $|R|$. $\square$

Binkowski et al. (2026) note that sink nodes are “likely to appear in the MST” of MTopDiv . Proposition 1 makes this exact, and it shows where the topology can enter: only through δ_P , which is non-zero only when a response token attends to an earlier response token more than one of the two attends to its best prompt token. We test the proposition on real heads, and TOHA against its first-order counterparts, in §5.4.

4 EXPERIMENTAL SETUP

Models and benchmarks. We use Qwen2.5-1.5B/3B/7B-Instruct, Phi-3-mini-4k-instruct, TinyLlama-1.1B-Chat, SmolLM-1.7B-Instruct, and Mistral-7B-Instruct-v0.2: 7 open models spanning 1.1B–7B parameters, with additional Mistral-7B analysis in Appendix G. **TruthfulQA** (Lin et al., 2022): all 817 questions, each paired with its best answer and first incorrect answer (1,634 rows) under each model’s chat format (Appendix B). **HaluEval QA** (Li et al., 2023a): the first 1,000 (Qwen2.5-3B) or 500 (Qwen2.5-1.5B) knowledge-grounded questions with right and hallucinated answers. Both are *teacher-forced*: the model reads dataset-written answers. **TriviaQA on-policy**: 2,000 random validation questions (Joshi et al., 2017); each model answers greedily (short-phrase

Table 1: **Theorem 1 on real attention.** *cells*: (example, layer) graphs; *viol.*: violations of any bound in Theorem 1(a)–(b) with $s = 0$; *coned*: share with $\delta_0 = 0$ exactly; $H_1 = \emptyset$ | *coned*: share of coned graphs with empty H_1 (C2 predicts 100%). Last two columns: median over layers of the Spearman correlation across examples between P_0 and S_0 (ρ_{raw}), and between $P_0/(N-1)$ and $1 - \bar{m} = S_0/(N-1)$ ($\rho_{/N}$), which removes the shared dependence on sequence length.

Benchmark	Model	$\bar{N}$	graphs	viol.	sink attn.	coned	$H_1 = \emptyset$ coned	ρ_{raw}	$\rho_{/N}$
TruthfulQA	Qwen2.5-3B	44.2	58,824	0	0.50	62.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-1.5B	44.2	45,752	0	0.53	64.5%	100.0%	1.000	1.000
TruthfulQA	Phi-3-mini	29.0	52,288	0	0.69	88.4%	100.0%	1.000	1.000
TruthfulQA	TinyLlama-1.1B	42.3	35,948	0	0.53	36.2%	100.0%	1.000	0.982
TruthfulQA	SmolLM-1.7B	32.4	39,216	0	0.02	0.0%	–	0.967	0.412
TruthfulQA	Mistral-7B	33.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Mistral-7B (chat)	34.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-7B	44.2	45,752	0	0.01	0.0%	–	0.976	0.513
HaluEval	Qwen2.5-3B	126.9	72,000	0	0.42	52.7%	100.0%	1.000	1.000
HaluEval	Qwen2.5-1.5B	128.1	28,000	0	0.45	62.5%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-3B	68.6	72,000	0	0.48	63.0%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-1.5B	68.8	56,000	0	0.52	50.6%	100.0%	1.000	1.000
TriviaQA (on-policy)	Phi-3-mini	46.1	64,000	0	0.69	87.1%	100.0%	1.000	1.000
TriviaQA (on-policy)	TinyLlama-1.1B	72.0	44,000	0	0.50	36.4%	100.0%	1.000	0.991
TriviaQA (on-policy)	Mistral-7B	58.7	64,000	0	0.64	65.8%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-7B	68.3	56,000	0	0.01	0.0%	–	0.986	0.489

instruction, ≤ 24 tokens), an answer is labelled hallucinated unless a normalized gold alias occurs in it, and features are extracted by re-reading the model’s *own* answer. Per-model accuracies are in Table 5. All models run in bfloat16 with eager attention.

Features. Per layer (head-averaged A): 0D (max lifetime, total persistence); 1D (four summaries, divided by N); SINK (S_0 and $\max_u D_{0u}$, i.e. exactly the two quantities Corollary 2 says 0D reduces to); DEFLATED PH (0D+1D after deleting token 0 and renormalizing rows); ANSWER-ONLY PH (induced subgraph on answer tokens); row entropy and row max. Per head: 0D (MST total and max via a dense Prim’s algorithm), sink mass, entropy, Lookback ratio (Chuang et al., 2024), and the LLM-Check attention score (Sriraman et al., 2024). TOHA (per head, separate forward pass, same rows): the exact score d of Bazarova et al. (2026) via a dense Prim’s algorithm on the prompt-contracted graph, the $\bar{\pi}$ -score, the response sink score, and δ_P (Proposition 1). Other: answer-token log-probability and predictive-entropy statistics; mean-pooled answer hidden states at the middle layer; TF-IDF word and character n -grams of the answer text (LEXICAL).

Protocol. Every number uses one evaluator: standardized features and L_2 logistic regression, with $C=1$ for banks of at most 64 dimensions and otherwise $C \in \{0.01, 0.1, 1\}$ chosen by inner grouped 3-fold CV. The outer loop is 10-fold StratifiedGroupKFold grouped by question, repeated over 3 seeds (42–44), with out-of-fold probabilities averaged. Incremental value is tested in two ways. *Early fusion* (“X|Y”) concatenates the banks. *Late fusion* (“X $\oplus$ Y”) stacks the two banks’ out-of-fold logits in a second-level logistic regression under the same CV, so that a small bank cannot be drowned out by a wide one. Differences in pooled AUC use a 10,000-resample question-level cluster bootstrap. We call two banks *equivalent* if the 90% interval lies inside ± 0.015 AUC (two one-sided tests, TOST, at $\alpha=0.05$), and we report TOST power. Paired benchmarks also report within-pair accuracy. The theory checks use every (example, layer) graph. Ripser (Bauer, 2021) stores filtration values in 32-bit floats, so bar-length and max-death bounds are checked at tolerance 10^{-5} , and the P_0 sandwich, a sum of up to N such values, at 10^{-3} .

5 RESULTS

5.1 THE REDUCTION HOLDS ON REAL ATTENTION

Table 1 checks Theorem 1 on every layer-level graph of every example: 838,356 graphs in total. There are 0 violations of any bound. In the 13 settings whose models place 0.42 to 0.75 of attention on token 0, 36–98% of graphs are *exactly* coned ($\delta_0 = 0$), and on every one of them H_1 is empty, as prediction C2 requires. Across graphs, δ_0 and 1D total persistence have Spearman correlation 0.07 to 0.89 (C3), the floor being the 2 settings above 90% coned, where both quantities are near-constant (99.99% of graphs have no 1D bars) and the correlation is degenerate rather than weak; elsewhere it

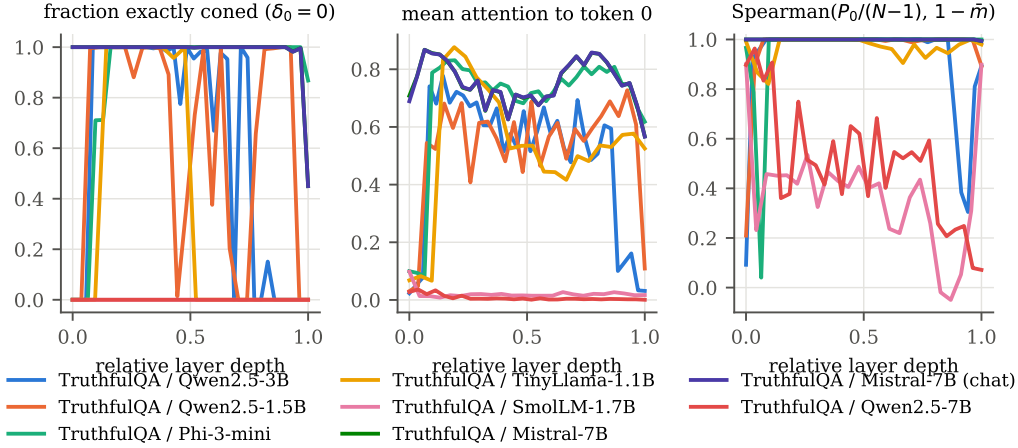


Figure 1: **Layerwise structure (TruthfulQA).** Coning (left) tracks sink attention (middle): layers with little sink attention, typically the first few and, in some models, the last few, are rarely coned, and the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B) are never coned at $s = 0$. Right: length-normalized rank correlation between P_0 and sink attention. It is near 1 in the layers of sink models, dips in some open layers, and is low throughout the two without a sink.

is at least 0.32. Both P_0 and the star weight $S_0 = (N-1)(1 - \bar{m})$ scale with sequence length, so their raw per-layer rank correlation (median ≥ 0.999) partly reflects length. After dividing both by $N-1$, the median per-layer rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ is still at least 0.98 in every sink setting (C1). In most layers, then, length-normalized 0D total persistence is, up to rank, one minus the mean attention paid to the first token. The weakest single layer per setting reaches only -0.14 to 0.99.

Figure 1 shows where coning happens: in the middle of the network, where sink attention is high. The first few layers are never sink-dominated, and in some models (TinyLlama, the Qwen models) so is a block of late layers; these layers are rarely coned. There, the length-normalized correlation can drop, as Theorem 1(b) allows once $(N-1)\delta_0$ is comparable to the spread of S_0 across examples.

Models without a first-token sink as a contrast. Our study includes 2 models that place almost no attention on token 0 — Qwen2.5-7B and SmolLM-1.7B, at 0.01 to 0.02 — and none of their graphs is coned at $s = 0$, so Theorem 1 at the first token gives no useful guarantee for them. They are not alike. Allowing any apex, 0% of SmolLM-1.7B’s graphs are coned — no vertex cones any of them — against 10% of Qwen2.5-7B’s, whose attention is strongly concentrated, just not on token 0 (§5.4): one model has no cone structure to find, the other is one where $s = 0$ is the wrong apex. Their H_1 is non-empty on 55% to 87% of graphs, and although the raw $\rho(P_0, S_0)$ is 0.967 to 0.986, this is almost entirely shared length: the length-normalized correlation is 0.41 to 0.51. Both differ from the sink-dominated models in weights, data and template, so this is a contrast, not a controlled intervention; the within-model intervention is deflation (C4, below). The reduction is a property of sink-dominated attention, not of the pipeline.

5.2 WHAT TOPOLOGY ADDS BEYOND THE SINK

If 0D persistence reduced *exactly* to the star, a classifier on the two sink features would match one on the two 0D features in every layer. Table 10 (“Sink vs. 0D”) shows the two banks differ by -0.015 to +0.031 AUC across the 13 sink settings evaluated, and are statistically equivalent within ± 0.015 in 8 of them (the TOST power of this comparison is modest, so non-equivalence is not evidence of a difference). Adding 0D to the sink bank gains 0.000 to +0.028.

The residual lives exactly where the theory says it can. We split layers without using labels: *coned* layers ($\geq 90\%$ of graphs coned) and *open* layers ($< 50\%$). Adding the coned layers’ 0D features to SINK changes AUC by -0.001 to +0.004, whereas the open layers account for +0.003 to

Table 2: **Pooled out-of-fold ROC-AUC** (3-seed repeated grouped 10-fold CV; seed SD ≤ 0.008). TQA: TruthfulQA; HE: HaluEval; TrQA: on-policy TriviaQA. Lex.: answer-text TF-IDF; LogP: log-probability statistics; LLM-C.: LLM-Check; Sink: $(S_0, \max_u D_{0u})$ per layer; 0D/1D: persistence per layer; subscript h : per head (Sink $_h$: sink mass; 0D $_h$: MST total and max); Lookb.: Lookback ratios; Hidden: mid-layer probe; Non-topo.: LogP+LLM-C.+Lookb.+Hidden+row statistics. Best per row in bold. Full table in Appendix E.

Setting	Lex.	LogP	LLM-C.	Sink	0D	1D	Sink $_h$	0D $_h$	Lookb.	Hidden	Non-topo.
TQA Qw3B	.820	.630	.693	.737	.736	.660	.851	.853	.897	.913	.926
TQA Qw1.5B	.820	.638	.666	.723	.726	.586	.830	.838	.851	.916	.917
TQA Phi-3	.820	.578	.653	.651	.646	.526	.871	.868	.892	.933	.936
TQA TinyLL	.820	.637	.614	.643	.654	.592	.808	.806	.850	.883	.885
TQA SmolLM	.820	.588	.604	.611	.583	.543	.720	.800	.863	.894	.899
TQA Mis7B	.820	.641	.652	.685	.684	.501	.903	.893	.933	.956	.959
TQA Mis7B (chat)	.820	.643	.653	.688	.688	.501	.907	.900	.930	.958	.960
TQA Qw7B	.820	.674	.667	.749	.679	.610	.821	.850	.922	.933	.950
HE Qw3B	.981	.993	.745	.829	.811	.710	.958	.971	.998	1.00	1.00
HE Qw1.5B	.979	.995	.791	.820	.789	.747	.922	.949	.999	1.00	1.00
TrQA Qw3B	.596	.839	.688	.717	.732	.601	.820	.807	.861	.841	.870
TrQA Qw1.5B	.609	.823	.676	.707	.716	.574	.767	.768	.824	.855	.873
TrQA Phi-3	.635	.877	.676	.707	.710	.546	.826	.807	.840	.867	.883
TrQA TinyLL	.740	.724	.688	.720	.720	.659	.814	.775	.823	.827	.842
TrQA Mis7B	.677	.719	.686	.696	.699	.611	.812	.775	.822	.833	.841
TrQA Qw7B	.594	.847	.725	.714	.668	.582	.801	.798	.879	.854	.881

+0.027 (Appendix E). Almost everything 0D adds beyond sink attention comes from the layers where the sink does not dominate.

Per head, the reduction is tighter. Per-head sink mass and per-head 0D are far stronger than their head-averaged versions (Table 2), but adding per-head MST total ($P_{0,h}$) to per-head sink mass changes AUC by only -0.001 to +0.011, which is equivalent within ± 0.015 in 13 of the 13 sink settings with per-head features.

The coning defect is a signal, but not a new one. Theorem 1 singles out δ_0 as the only place where topology can depart from the sink, so we also test it as a detector. Head-averaged, it is weak (AUC 0.50 to 0.67). Per head it is strong (0.69 to 0.96), and stacked onto per-head sink mass it adds +0.004 to +0.029 AUC (95% interval above 0 in 16 of 16 settings). Stacked onto the non-topological probes, however, it adds only 0.000 to +0.009 (above 0 in 5, equivalent within ± 0.015 in 16 of 16).

Deleting the sink exposes some, redundant, structure. Deflated PH (C4) adds +0.029 to +0.067 over SINK on TruthfulQA (95% interval excludes 0 in 6 of 6 sink settings) but only +0.002 to +0.043 on the on-policy benchmark (2 of 5). Answer-only PH behaves similarly (Appendix E). So attention graphs carry some non-sink geometry, most clearly on teacher-forced text. As §5.3 shows, this geometry is already available to non-topological probes.

One-dimensional persistence. Most real graphs have no 1D bars at all (Table 1), unlike i.i.d. weights, where 1D persistence grows linearly. Adding 1D to 0D changes AUC by -0.008 to +0.033 and is equivalent within ± 0.015 in 9 of 13 sink settings. A sink dose-response experiment and a planted-cycle experiment (Appendix D) show why: under a strong sink, a loop is visible to H_1 only if it breaks the cone.

5.3 HALLUCINATION DETECTION AGAINST STRONG BASELINES

Table 2 places attention-graph topology among standard detectors. Three patterns hold across benchmarks.

Topology is dominated by cheap non-topological probes. On TruthfulQA and the on-policy benchmark, a mean-pooled hidden-state probe (0.83 to 0.96) and per-head Lookback ratios (0.82 to 0.93) outperform every topological bank, and their combination with log-probability, LLM-Check, and row

Table 3: **TOHA against its first-order counterparts** (Proposition 1). $\delta_P=0$: share of (example, head) graphs that are exactly prompt-coned, on which d equals the $\bar{\pi}$ -score. $\rho(d, \bar{\pi})$: median over heads of the rank correlation across examples. AUC columns: TOHA’s head-selection algorithm run on d (TOHA), on the $\bar{\pi}$ -score (TOHA $[\bar{\pi}]$), and on the response sink score (TOHA $[\bar{a}_0]$); then supervised probes on all heads of each score; then the non-topological bank. Δ columns and marks as in Table 10; $\oplus$ is late fusion.

Setting	$\delta_P=0$	$\rho(d, \bar{\pi})$	TOHA	TOHA $[\bar{\pi}]$	TOHA $[\bar{a}_0]$	TOHA $_{\text{sup}}$	$\bar{\pi}_{\text{sup}}$	$\bar{a}_0_{\text{sup}}$	Non-topo.	$\bar{\pi}$ -TOHA	$\bar{a}_0$ -TOHA	TOHA $\oplus\bar{\pi}$	TOHA $\oplus\text{NT}$
TQA Qw3B	57%	.976	.668	.670	.649	.892	.892	.892	.926	+0.003 $\equiv$	-0.018*	+0.003 $\equiv$	+0.003 $\equiv$
TQA Qw1.5B	58%	.978	.646	.649	.628	.858	.857	.857	.917	+0.003 $\equiv$	-0.018*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Phi-3	74%	.996	.751	.742	.715	.898	.901	.905	.936	-0.009*	-0.036*	0.000 $\equiv$	0.000 $\equiv$
TQA TinyLL	74%	.995	.666	.655	.618	.845	.843	.848	.885	-0.012*	-0.049*	+0.002 $\equiv$	+0.001 $\equiv$
TQA SmoLLM	58%	.954	.701	.695	.595	.862	.860	.841	.899	-0.007 $\equiv$	-0.107*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Mis7B	81%	.998	.751	.764	.766	.932	.934	.936	.959	+0.013*	+0.015*	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw7B	57%	.980	.705	.709	.544	.925	.927	.906	.950	+0.004 $\equiv$	-0.161*	+0.001 $\equiv$	+0.004 $\equiv$
HE Qw3B	68%	.987	.967	.982	.961	.999	.999	1.00	1.00	+0.015*	-0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	69%	.988	.970	.988	.972	.999	.999	.999	1.00	+0.018*	+0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	76%	.991	.728	.736	.742	.862	.863	.870	.870	+0.009	+0.014*	+0.002 $\equiv$	+0.012*
TrQA Qw1.5B	77%	.992	.712	.714	.708	.828	.827	.836	.873	+0.002 $\equiv$	-0.005	+0.003 $\equiv$	+0.004 $\equiv$
TrQA Phi-3	88%	.998	.724	.731	.733	.847	.851	.850	.883	+0.007 $\equiv$	+0.008	0.000 $\equiv$	+0.001 $\equiv$
TrQA TinyLL	66%	.994	.714	.725	.693	.833	.826	.832	.842	+0.011*	-0.021*	+0.007 $\equiv$	+0.010*
TrQA Mis7B	75%	.993	.716	.706	.710	.818	.820	.817	.841	-0.010	-0.007	+0.002 $\equiv$	+0.008 $\equiv$
TrQA Qw7B	76%	.994	.754	.790	.617	.876	.880	.842	.881	+0.036*	-0.137*	+0.001 $\equiv$	+0.012*

statistics reaches 0.84 to 0.96. Adding 0D PH to that combination changes AUC by 0.000 to +0.001 (equivalent within ± 0.015 in 16 of the 16 settings with per-head features), and adding sink-deflated PH changes it by -0.001 to +0.002 (equivalent in 16 of 16). Early fusion could hide a small bank inside a $\sim 2,000$ -dimensional one, so we repeat the test with late fusion. Stacking the 0D score onto the non-topological score changes AUC by 0.000 to +0.003 (equivalent in 16 of 16 settings; the 95% interval lies above 0 in 0 of them), and stacking the deflated-PH score changes it by 0.000 to +0.006 (equivalent in 16 of 16).

Which cheap probe wins depends on whether text is teacher-forced. On TruthfulQA, token log-probability is weak (0.58 to 0.67), consistent with the benchmark’s design around imitative falsehoods, and 0D PH (0.58 to 0.74) scores higher for 7 of 8 models, significantly so for 5. On-policy, log-probability is strong (0.72 to 0.88) and significantly beats 0D (0.67 to 0.73) for 4 of 6 models. Conclusions about topology versus confidence drawn from a single teacher-forced benchmark do not transfer.

Per-head resolution matters more than topology. Per-head features of every kind are far stronger than head-averaged ones, and per-head entropy lands within -0.016 to +0.044 AUC of per-head 0D (Appendix E).

5.4 THE PUBLISHED TOPOLOGICAL DETECTOR IS FIRST-ORDER

We test Proposition 1 on every head of every example: 18,906,944 head graphs from 15 settings (Table 3). We then run TOHA’s head-selection algorithm (Bazarova et al., 2026, Alg. 1) inside our cross-validation, selecting heads on the whole training fold, which is larger than TOHA’s probe set. We run it once on TOHA’s score d and, unchanged, on the two first-order scores the proposition names: the $\bar{\pi}$ -score $\frac{1}{|R|} \sum_u (1 - \pi_u)$ and the response sink score $1 - \bar{a}_0 = \frac{1}{|R|} \sum_u (1 - A_{u0})$.

The identity holds on real heads. The sandwich has 0 violations. The score d is our own implementation, so we also check it against the authors’ released code: on 21,696 head graphs from 2 settings, their MTOP-DIV routine (Bazarova et al., 2026), which uses ripser rather than our dense Prim’s algorithm, agrees with ours to 1.2×10^{-8} . Per setting, 57–88% of head graphs are exactly prompt-coned, and on those d equals the $\bar{\pi}$ -score up to float rounding (largest gap $6 \cdot 10^{-8}$). Across examples, the median per-head rank correlation between d and the $\bar{\pi}$ -score is 0.954 to 0.998. With the sink score it is 0.78 to 0.99 in the sink models, but -0.16 to -0.09 in Qwen2.5-7B and SmoLLM-1.7B, which have no first-token sink: there d tracks the largest prompt attention just as tightly while being, if anything, mildly anti-correlated with attention to token 0. The reduction is to max-prompt attention; that this coincides with the sink is a property of the model, not of the theorem. Shifting the sink’s attention logit moves these quantities as predicted (Appendix F).

Topology adds almost nothing to the first-order score. Stacking d onto the $\bar{\pi}$ -score changes AUC by 0.000 to +0.007. The change is equivalent to zero within ± 0.015 in 15 of 15 settings, although the

95% interval lies above 0 in 4 of them. With supervised probes on all heads (the setting of TOHA’s own feature comparison), the $\bar{\pi}$ -score differs from d by -0.007 to +0.004 (equivalent in 15 of 15). The response sink score differs from d by -0.034 to +0.008 (equivalent in 11). The exceptions are the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B), where the sink score is worse, and one on-policy setting whose interval extends past the margin on the positive side. TOHA’s discrete head selection is noisier: run on the $\bar{\pi}$ -score it changes AUC by -0.012 to +0.036 (significantly higher in 5 settings, lower in 2, non-inferior at margin 0.015 in 12 of 15). Run on the sink score it loses up to 0.049 AUC in the sink models (0.107 to 0.161 in Qwen2.5-7B and SmolLM-1.7B, which have no first-token sink): the selected heads are prompt-coned on 57–95% of examples, but the sink is every response token’s top prompt token on only 19–79%. Stacked onto the non-topological probes, TOHA changes AUC by 0.000 to +0.012 (95% interval above 0 in 6 of 15); by the above, whatever it adds is carried by the $\bar{\pi}$ -score.

Why “attention to $\langle s \rangle$ ” looked weak. Bazarova et al. (2026, Table 9) report that attention to the first token is a much weaker feature than their divergence. Proposition 1 identifies the quantity that d actually tracks: per head, the attention of the *response* tokens to their most-attended prompt token. In the sink models, that token is the sink for every response token in 40–83% of head graphs.

Benchmark artifacts. A TF-IDF model on the answer text alone reaches 0.82 AUC on TruthfulQA and 0.98 on HaluEval, but only 0.59 to 0.74 on-policy: teacher-forced benchmarks reward recognizing how the dataset’s answers were *written* (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). We draw no conclusions from HaluEval, where lexical separability is near perfect.

6 DISCUSSION

What attention-graph topology measures. In decoder-only models, Vietoris–Rips persistence of an attention graph, and every stable summary of it, is controlled by one scalar per graph: the coning defect of the first token. The same is true of TOHA’s prompt–response divergence, with the prompt playing the role of the apex. Where it is zero, as in most middle-layer graphs of every sink-dominated model we tested, 0D total persistence equals $(N-1)(1-\bar{m})$, a function of length and sink attention alone, and higher homology is empty. Where it is positive, Theorem 1 bounds how far topology can depart from the sink; in our data this happens mainly in the first layers.

Recommendations. (i) Report sink attention and δ_0 (or the relevant defect, such as δ_P for TOHA) next to any attention-graph topological feature, and compare the detector with the first-order statistic that the reduction names. (ii) Compare topological detectors against per-head non-topological statistics (sink mass, entropy, Lookback ratios) at the same resolution, and against a hidden-state probe. (iii) Evaluate on the model’s own generations; teacher-forced benchmarks mix detection with recognition of dataset authorship. (iv) If structure beyond the sink is the goal, remove the sink first (deflation) or restrict to answer tokens. Both expose real non-sink geometry, but in our settings it was not new information.

Limitations. Our models range over 1.1B–7B parameters; sink strength changes with scale and training (Gu et al., 2025), so the *share* of coned layers is model-specific, although Theorem 1 is not. Our results cover Vietoris–Rips persistence of symmetrized attention (head-averaged or per head), its causal directed-flag equivalent, every stable vectorization of it, and TOHA. They do not cover HalluZig (Samaga et al., 2026), which computes zigzag persistence of the *graph* (1-skeleton) homology of top-percentile edge sets across layers: a cone graph still has cycles, because its triangles are not filled, so Theorem 1 does not constrain its H_1 . Learned graph models (Frasca et al., 2026) are also outside the theorem, although their inputs are the same sink-dominated graphs. TOHA was designed for RAG; only HaluEval supplies a context passage here. Our on-policy labels come from alias string matching, which mislabels some paraphrases and hedged answers. An LLM-judge audit of 200 answers in 2 on-policy settings finds 6% to 14% disagreement; re-scoring those rows under the judge’s labels shifts every bank by at most 0.058 AUC and leaves the dominance orderings intact. The judge is itself a model, so the labels remain unaudited against human judgment (Janiak et al., 2025). TinyLlama’s on-policy answers were generated without the BOS token added at feature extraction. Our intervals resample questions for a fixed, seed-averaged predictor and do not include variation

from refitting, so equivalence claims are conditional on the trained models. Prompt formats differ across models (Appendix B), so cross-model comparisons mix architecture and template.

REPRODUCIBILITY STATEMENT

All code is released at <https://anonymous.4open.science/r/ICLR-BE07/>. `sinktda/extract.py` computes all whole-graph and per-head features, `sinktda/toha.py` computes TOHA and its first-order counterparts and runs TOHA’s head selection, and `sinktda/evaluate.py` produces every AUC, interval and theory check. `sinktda/late_fusion.py` and `sinktda/defect_probe.py` run the stacked tests, `sinktda/check_theory.py` and `sinktda/synthetic.py` run the numerical and synthetic checks, and `sinktda/report.py` generates every table, figure and number in the text. The run order is in `sinktda/run_fix.sh` and `sinktda/run_toha.sh`. Per-example features for all settings and the TriviaQA generations are included. Seeds, prompts, dtype and model identifiers are listed in Appendix B; proofs are in Appendix A.

LLM USAGE STATEMENT

LLMs were used to help implement experiment code, check reported statistics against result files, and draft and edit text. The authors verified every claim, number, proof, and citation and take full responsibility for the paper.

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A PROOFS

Throughout, D is a symmetric $N \times N$ matrix with non-negative entries and zero diagonal, K_ϵ is the Vietoris–Rips complex at scale ϵ , $\text{filt}_D(\sigma) = \max_{u,v \in \sigma} D_{uv}$ (with $\text{filt}_D(\{u\}) = 0$), and homology is taken over a field. Bars are half-open intervals $[b, d)$.

A.1 FACT 1

All vertices enter at 0. Processing edges in non-decreasing weight order, an edge either joins two components, in which case (elder rule) one 0D class dies at its weight and Kruskal’s algorithm adds the edge to the spanning forest, or it closes a cycle, in which case no 0D class dies and Kruskal discards it. Hence the $N - 1$ finite death times are the MST edge weights (Gower & Ross, 1969; Edelsbrunner & Harer, 2010).

A.2 THEOREM 1

Lemma 1 (Coned filtrations). *If D is coned at s , then for every $\epsilon \geq 0$, $K_\epsilon = C_\epsilon \sqcup I_\epsilon$, where C_ϵ is a simplicial cone with apex s and I_ϵ is a set of isolated vertices. Consequently $\tilde{H}_k(K_\epsilon) = 0$ for every $k \geq 1$.*

Proof. Let $\sigma \in K_\epsilon$ with $s \notin \sigma$ and $|\sigma| \geq 2$. For each $u \in \sigma$ choose $v \in \sigma \setminus \{u\}$; coning gives $D_{su} \leq D_{uv} \leq \text{filt}_D(\sigma) \leq \epsilon$. Therefore $\text{filt}_D(\sigma \cup \{s\}) = \text{filt}_D(\sigma)$ and $\sigma \cup \{s\} \in K_\epsilon$. Now let $u \neq s$ be a vertex with $D_{su} > \epsilon$. If u had an edge $\{u, v\} \in K_\epsilon$ then $D_{su} \leq D_{uv} \leq \epsilon$, a contradiction; so u is isolated. Let I_ϵ be these vertices and C_ϵ the subcomplex on the remaining vertices. Every vertex of C_ϵ is adjacent to s , and by the first part every simplex of C_ϵ extends by s , so C_ϵ is the cone over its link of s and is contractible. H_k of a disjoint union is the direct sum, and isolated points have no homology in degree ≥ 1 . $\square$

Proof of Theorem 1. Fix s and $\delta = \delta_s(D)$. Define $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$, $u \neq v$; $D'_{su} = D'_{us} = D_{su}$; $D'_{uu} = 0$.

Step 1: $D \leq D' \leq D + \delta$ and D' is coned at s . The first inequality is immediate. For $u, v \neq s$, $D'_{uv} - D_{uv} = [\max(D_{su}, D_{sv}) - D_{uv}]_+ \leq \delta$. Coning: $\max(D'_{su}, D'_{sv}) = \max(D_{su}, D_{sv}) \leq D'_{uv}$.

Step 2: interleaving. Since filtration values are maxima of entries, $\text{filt}_D \leq \text{filt}_{D'} \leq \text{filt}_D + \delta$. Hence for every b , $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$, where K' is the filtration of D' .

Step 3: (a). The persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ is induced by the composite inclusion in Step 2 and therefore factors through $H_k(K'_{b+\delta})$, which is zero for $k \geq 1$ by Lemma 1. The rank of $H_k(K_b) \rightarrow H_k(K_{b'})$ equals the number of bars $[b_i, d_i)$ with $b_i \leq b$ and $d_i > b'$. A bar with $d_i - b_i > \delta$ would contribute to this rank at $b = b_i$, $b' = b_i + \delta$, a contradiction. So every bar has length at most δ , and summing gives the bound on P_k .

Step 4: (b). By Fact 1, $P_0(D) = w_D(T_D)$ for an MST T_D of D . The star $T_s = \{\{s, u\}\}_{u \neq s}$ is a spanning tree, so $P_0(D) \leq w_D(T_s) = S_s(D)$. For the lower bound, T_s is an MST of D' , even with ties. Root any spanning tree T at s and let e_u be the edge from $u \neq s$ to its parent $p(u)$. If $p(u) = s$ then $D'_{e_u} = D'_{su}$; otherwise $D'_{e_u} = D'_{up(u)} \geq \max(D'_{su}, D'_{sp(u)}) \geq D'_{su}$ because D' is coned. The map $u \mapsto e_u$ is a bijection onto the edges of T , so $w_{D'}(T) \geq \sum_{u \neq s} D'_{su} = w_{D'}(T_s)$ and $\max_{e \in T} D'_e \geq \max_u D'_{su}$. Hence $w_{D'}(T_{D'}) = w_{D'}(T_s) = S_s(D)$, and $P_0(D) = w_D(T_D) \geq w_{D'}(T_D) - (N-1)\delta \geq w_{D'}(T_{D'}) - (N-1)\delta = S_s(D) - (N-1)\delta$. The maximum finite 0D death is the bottleneck value $\min_T \max_{e \in T} w(e)$, which is attained by any MST; it is at most $\max_u D_{su}$ (the star is a spanning tree) and is 1-Lipschitz in the sup-norm, which gives the lower bound because the bottleneck of D' is $\max_u D_{su}$ (by the rooted-tree argument above).

Relation to known results. Lemma 1 is the classical observation that a flag complex in which one vertex is adjacent to every non-isolated vertex, and to every simplex, is a cone; this is the dominated-vertex mechanism behind strong collapses (Barmak & Minian, 2012) and edge collapses of flag filtrations (Boissonnat & Pritam, 2020). Step 3 is a one-sided instance of the interleaving argument behind the stability of persistence (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). The

two-sided stability theorem would give bars of length at most 2δ , whereas the one-sided comparison $D \leq D'$ gives δ , which is attained (Section 3).

Step 5: (c). If $\delta = 0$ then $D' = D$, and the claims follow from Lemma 1 and Step 4. $\square$

Proof of Corollary 1. Let D' be as in the proof of Theorem 1. Since D' is coned at s , Lemma 1 and Step 4 give $\text{dgm}_k(D') = \Sigma_s(D)_k$ for all k : no bars in degree $k \geq 1$, and 0D deaths equal to the star weights $D'_{su} = D_{su}$. For $1 \leq j \leq N-1$, the j -th smallest finite 0D death of a matrix E is $w_j(E) = \min\{\epsilon : \beta_0(K_\epsilon^E) \leq N-j\}$. If $E \leq E'$ entrywise then $K_\epsilon^{E'} \subseteq K_\epsilon^E$ on the same vertex set, so $\beta_0(K_\epsilon^E) \leq \beta_0(K_\epsilon^{E'})$ and $w_j(E) \leq w_j(E')$; and $w_j(E+c) = w_j(E) + c$ on off-diagonal entries. From $D \leq D' \leq D + \delta_s$ we get $w_j(D) \leq D_{s(j)} \leq w_j(D) + \delta_s$. The same inclusions give the Betti-0 sandwich, because $\beta_0(K_\epsilon^D) = 1 + \#\{u : D_{su} > \epsilon\}$. Matching the j -th deaths shows that $d_B(\text{dgm}_0(D), \Sigma_s(D)_0) \leq \delta_s$ and $W_1 \leq (N-1)\delta_s$. In degree $k \geq 1$, every bar has length at most δ_s (Theorem 1(a)), so matching it to the diagonal costs at most $\delta_s/2$ in ℓ^∞ . The Lipschitz statements follow from the stability of landscapes in d_B (Bubenik, 2015) and of persistence images in W_1 (Adams et al., 2017). If $\delta_s = 0$ then $D = D'$. $\square$

Proof of Proposition 1. Let $\tilde{D}$ equal D except that $\tilde{D}_{vv'} = 0$ for $v, v' \in P$. At scale 0, Kruskal's algorithm joins all of P into one component with $|P| - 1$ zero-length bars. Every later merge is a merge in the quotient graph Q on $\{p^*\} \cup R$ with $Q_{p^*u} = \min_{v \in P} \tilde{D}_{vu}$ and $Q_{uw} = D_{uw}$. So $\text{MTopDiv}(R, P) = P_0(Q)$. For $u \in R$ and $v \in P$ we have $v < u$ and $D_{uv} = 1 - A_{uv}$, so $Q_{p^*u} = 1 - \pi_u$. For $w < u$ in R , $Q_{uw} = 1 - A_{uw}$, and $\max(Q_{p^*u}, Q_{p^*w}) - Q_{uw} = A_{uw} - \min(\pi_u, \pi_w)$, so $\delta_{p^*}(Q) = \delta_P$. Theorem 1(b) with $N-1 = |R|$ gives the sandwich, with equality when $\delta_P = 0$ by part (c). Finally $\pi_u \geq A_{u0}$ because $0 \in P$. $\square$

We checked both statements numerically. The fast MTop-Div (a dense Prim's algorithm on Q) agrees with ripser run on $\tilde{D}$ to $3 \cdot 10^{-8}$ on 400 random causal matrices. On these, and on 300 further random matrices checked for the elementwise death bound and the bar bound in degrees 1 and 2, there are 0 violations (sinktda/check_theory.py).

Proof of Corollary 2. For $u > v$, $A_{vu} = 0$, so $D_{uv} = 1 - A_{uv}$; in particular $D_{0u} = 1 - A_{u0}$. For $1 \leq v < u$, $\max(D_{0u}, D_{0v}) - D_{uv} = A_{uv} - \min(A_{u0}, A_{v0})$. Substitute into Theorem 1. $\square$

A.3 LINEAR GROWTH OF 1D PERSISTENCE UNDER I.I.D. WEIGHTS

Length normalization of 1D features is usually motivated by the growth of cycles with N . For i.i.d. weights this growth is known: 0D total persistence converges to $\zeta(3)$ (Frieze, 1985), and 1D total persistence of the random clique-complex process is $\Theta(N)$ (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019). The following elementary bound makes the constant explicit.

Proposition 2 (Linear growth under i.i.d. weights). *If $\{D_{uv}\}_{u < v}$ are i.i.d. Uniform $[0, 1]$, then $\mathbb{E}[P_1(N)] \geq \int_0^1 \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+ dp = \frac{3}{8}N - \sqrt{3N} + O(1)$. (A matching $O(N)$ upper bound is Theorem 1.5 of Hino & Kanazawa, 2019.)*

Proof. With i.i.d. uniform weights, the edges with $D_{uv} \leq p$ form an Erdős-Rényi graph $G(N, p)$ and K_p is its clique (flag) complex. Every 1D bar is contained in $[0, 1]$, because K_1 is the full simplex. Writing $\beta_1(p)$ for the first Betti number of K_p , the number of bars alive at p is $\beta_1(p)$, so $P_1 = \int_0^1 \beta_1(p) dp$. For any finite complex with f_i faces of dimension i , the strong Morse inequality at degree 2 reads $\beta_2 - \beta_1 + \beta_0 \leq f_2 - f_1 + f_0$ (Edelsbrunner & Harer, 2010), hence $\beta_1 \geq f_1 - f_0 - f_2 + \beta_0 + \beta_2 \geq f_1 - f_0 - f_2$. Since also $\beta_1 \geq 0$, Jensen's inequality gives $\mathbb{E}\beta_1(p) \geq [\mathbb{E}f_1 - f_0 - \mathbb{E}f_2]_+ = \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+$. Integrating over p gives the first inequality. With $p = x/\sqrt{N}$ and $dp = dx/\sqrt{N}$, the integrand divided by $\sqrt{N}$ is $N(x/2 - x^3/6) - \sqrt{N} + O(1)$ uniformly for $x \in [0, \sqrt{3} + 1]$. It is positive on an interval whose endpoints are $O(N^{-1/2})$ from 0 and $\sqrt{3}$, and it is negative for larger x . Hence $\int_0^1 [\cdot]_+ dp = N \int_0^{\sqrt{3}} \left(\frac{x}{2} - \frac{x^3}{6} \right) dx - \sqrt{3N} + O(1) = \frac{3}{8}N - \sqrt{3N} + O(1)$. Numerically, $(\frac{3}{8}N - \text{bound})/\sqrt{N} = 1.62, 1.67, 1.70, 1.72$ at $N = 2^8, 2^{10}, 2^{12}, 2^{14}$,

approaching $\sqrt{3} \approx 1.73$. The order $\Theta(N)$ itself is known (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); the proposition only makes the constant explicit. $\square$

Numerically, the integral equals 0.791, 3.92, 11.9, 30.2, 70.1 at $N = 16, 32, 64, 128, 256$, against simulated means 1.91, 5.83, 15.5, 36.2, 79.3 under i.i.d. Uniform $[0, 1]$ weights (Figure 4, left). Over $N \in [16, 256]$ both grow faster than linearly (log-log slopes 1.55 and 1.31), a pre-asymptotic effect of the $-\sqrt{3N}$ term; both are $\Theta(N)$ as $N \rightarrow \infty$.

A.4 DIRECTED FLAG COMPLEXES ON CAUSAL ATTENTION

Proposition 3 (Directed flag collapse). *Let A be lower-triangular with non-negative entries (self-loops are ignored). Let $\text{DFl}(A)$ be the filtered directed flag complex (Lütgehetmann et al., 2020) of the digraph with an edge $i \rightarrow j$ for every $j < i$ (a transitive tournament) and weight $1 - A_{ij}$, and let $\text{VR}(D)$ be as above with $D = 1 - \max(A, A^\top)$. Then $\text{DFl}(A)$ and $\text{VR}(D)$ are isomorphic filtered complexes, so $\text{dgm}_k(\text{DFl}(A)) = \text{dgm}_k(\text{VR}(D))$ for all k . The same holds for any weighting $w(i \rightarrow j) = \phi(A_{ij})$ with $D_{ij} = \phi(\max(A_{ij}, A_{ji}))$ and ϕ non-increasing.*

Proof. A directed k -simplex is an ordered tuple $(v_0, \dots, v_k)$ with an edge $v_a \rightarrow v_b$ for all $a < b$. Edges exist only from later to earlier tokens, so a vertex set S supports exactly one directed simplex, its elements in decreasing order; every other ordering uses a missing forward edge. (A missing edge is not a zero-weight edge; encoding it as weight 0 would admit all orderings, a common implementation error.) This bijection preserves faces. Its filtration value is $\max_{a < b} (1 - A_{v_a v_b})$, and for $u > v$, $D_{uv} = 1 - A_{uv}$, so the undirected value $\max_{u, v \in S} D_{uv}$ is the same number. $\square$

We verified the bijection and the filtration identity by explicit enumeration on 420 random causal matrices ($N \leq 8$, four families including tie-dense and underflowing matrices; 30,060 vertex sets; zero violations), and confirmed that the identity fails for bidirectional attention, where a set of size n supports $n!$ orderings. The collapse is a consequence of causality: for decoders, the directed flag complex adds nothing to its symmetrized counterpart. Directional features that do not come from flag complexes, such as those of Kushnareva et al. (2021) for bidirectional encoders, are outside this statement.

B IMPLEMENTATION AND PROTOCOL DETAILS

Extraction. All reported settings run in bfloat16 on Apple-silicon (MPS) with eager attention; float16 on MPS produced non-finite values and degenerate generations for Qwen models, and the extractor refuses non-finite logits. Model identifiers are in Table 6 (Hugging Face main revisions as of September 2026). For each example we run one forward pass with attentions and hidden states, average heads, and form D in float64. Ripser (Bauer, 2021) computes 0D/1D barcodes; per-head MSTs use a batched dense Prim’s algorithm (SciPy’s sparse MST drops explicit zero-weight edges, which occur when $A_{u0} \approx 1$). δ_0 uses the closed form of Corollary 2; $\min_s \delta_s$ is computed by brute force over all apexes. DEFLATED deletes token 0 and renormalizes each row over the remaining causal support. ANSWER-ONLY restricts D to the answer tokens without renormalization. The Lookback ratio of head h is the mean over answer tokens t of $\bar{a}_{\text{ctx}}/(\bar{a}_{\text{ctx}} + \bar{a}_{\text{new}})$, with $\bar{a}$ the mean attention from t to prompt tokens and to answer tokens up to t . The LLM-Check score of a head is the mean log diagonal attention (the log-determinant of the triangular attention kernel divided by N), summed over heads per layer. Log-probability features are the mean, minimum, and sum of answer-token log-probabilities, mean and maximum predictive entropy, and mean maximum softmax probability. Hidden states are mean-pooled over answer tokens at layer $\lfloor L/2 \rfloor$.

Prompts. TruthfulQA uses each model’s own format: the Qwen chat template with an explicit system turn “You are a helpful assistant.”, `<|user|>...<|end|>` for Phi-3, `<|user|>...</s>` for TinyLlama, ChatML without system turn for SmoLLM, and `[INST]...[/INST]` for Mistral. HaluEval uses Knowledge: ... \nQuestion: ... \nAnswer: without a chat template. The prompt boundary is computed by tokenizing the prefix without its trailing space, which the full string merges into the first answer token. The on-policy prompt is the model’s chat template around “Answer the following question with a short phrase only, no explanation. Question: ...”.

Token 0 is whatever token the format places first (BOS for TinyLlama and Mistral, `<|user|>` for Phi-3, whose tokenizer adds no BOS, `<|im_start|>` for Qwen and SmoLLM, and “Knowledge” for Qwen on HaluEval); the theorem does not require it to be a special token. On-policy answers were generated from the chat-formatted prompt with `add_special_tokens=False`, and features were extracted with the tokenizer’s default special tokens; the two differ only for TinyLlama, whose extraction input starts with a BOS token.

Sensitivity to inference precision. An earlier run of five settings in float16 lets us measure how much each bank depends on numerical precision (Table 4). Switching to bfloat16 changes the AUC of head-averaged topological banks by up to 0.058 (TQA Qw3B; at most 0.005 elsewhere), and of sink features by up to 0.003, but that of non-topological banks by at most 0.001. Short 1D bars live near the scale of δ_0 and are sensitive to small perturbations of attention, which is one more reason to report the precision alongside any attention-topology result.

Table 4: **AUC change from float16 to bfloat16** (same prompts and protocol; HaluEval Qwen2.5-3B also changed its prompt boundary, see above).

Setting	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw3B	−0.003	−0.018	+0.058	−0.001	−0.001	−0.001	0.000	0.000	−0.001
TQA Phi-3	−0.001	−0.001	−0.003	0.000	−0.001	0.000	0.000	0.000	0.000
TQA TinyLL	0.000	0.000	−0.001	−0.001	0.000	0.000	0.000	0.000	0.000
TQA SmoLLM	0.000	−0.001	−0.001	+0.001	+0.002	0.000	0.000	0.000	0.000
HE Qw3B	+0.001	−0.005	+0.001	+0.002	−	−0.001	−	0.000	−

On-policy benchmark. Table 5 summarizes the generations each model produced and how they were graded.

Table 5: **TriviaQA on-policy benchmark.** 2,000 validation questions sampled with seed 0; greedy decoding, first line of the answer, at most 24 new tokens. An answer is correct if any normalized gold alias (lower-cased, punctuation and articles removed) occurs in the normalized answer as a whole-word span. Empty answers are dropped.

Model	questions	accuracy	rows used	mean answer tokens
Qwen2.5-3B	2,000	42.4%	2,000	5.5
Qwen2.5-1.5B	2,000	31.4%	2,000	5.8
Phi-3-mini	2,000	56.2%	2,000	6.0
TinyLlama-1.1B	2,000	26.4%	2,000	18.9
Mistral-7B	2,000	66.5%	2,000	13.0
Qwen2.5-7B	2,000	51.4%	2,000	5.3

Statistics. Bootstrap resamples are drawn over questions (both rows of a pair move together). TOST power at margin m assumes a true difference of 0 and a normal sampling distribution with the bootstrap SE: $2\Phi(m/\text{SE} - z_{0.95}) - 1$. Repeated-CV predictions (seeds 42, 43, 44) are averaged before bootstrapping (seed 0), so the intervals reflect sampling of questions for a fixed, split-averaged predictor and omit refitting variance. The seed-to-seed SD of the pooled AUC is reported separately (Appendix E); it is at most 0.008 and exceeds 0.004 in 22 of the bank–setting cells. The C rule depends on bank width, so the same bank can be fitted with $C=1$ for a shallow model and with tuned C for a deeper one. Late fusion fits the second-level model on first-level out-of-fold scores, as in standard stacking; its training rows’ first-level scores come from models that saw the corresponding test questions, which is a mild, standard optimism shared by both arms of the comparison.

C ADDITIONAL ANALYSES

Models. Table 6 lists the audited architectures. All models use grouped or multi-head attention; we average attention over query heads.

Table 6: Audited models.

Model	params	layers	heads	KV heads	hidden
Qwen/Qwen2.5-3B-Instruct	3.09B	36	16	2	2048
Qwen/Qwen2.5-1.5B-Instruct	1.54B	28	12	2	1536
Microsoft/Phi-3-mini-4k-instruct	3.82B	32	32	32	3072
TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.10B	22	32	4	2048
HuggingFaceTB/SmolLM-1.7B-Instruct	1.71B	24	32	32	2048
mistralai/Mistral-7B-Instruct-v0.2	7.24B	32	32	8	4096

Length scaling of 1D persistence on real attention. Table 7 regresses log total 1D persistence (summed over layers) on $\log N$ with a class fixed effect, on rows with $P_1 > 0$, so that the exponent is not driven by a length difference between correct and hallucinated answers. Exponents range from 0.03 to 1.39, against 1.31 for i.i.d. uniform weights and 1.27 for random causal attention without a sink over the same range of N . 10 of 11 settings grow significantly more slowly than the i.i.d. reference (not HE Qw3B). The exponent exceeds 1 in HE Qw3B, HE Qw1.5B, so growth on real attention is not uniformly sub-linear. Summed over layers, almost every example has some 1D bar, because every model has some layers that are not coned; within an example, the coned layers contribute nothing to P_1 whatever the length (Table 1).

Table 7: Length scaling of 1D persistence. OLS of $\log P_1$ on $\log N$ and the class label; rows with $P_1 = 0$ are excluded and counted in the first column.

Setting	rows with $P_1 > 0$	$\hat{\alpha}$ (class FE)	95% CI	R^2	Spearman(P_1, N)
TQA Qw3B	100.0%	0.25	[0.23, 0.27]	0.27	0.57
TQA Qw1.5B	100.0%	0.57	[0.52, 0.61]	0.29	0.51
TQA Phi-3	99.9%	0.93	[0.82, 1.04]	0.15	0.39
TQA TinyLL	100.0%	0.32	[0.28, 0.36]	0.14	0.28
TQA SmolLM	100.0%	0.61	[0.54, 0.69]	0.13	0.36
HE Qw3B	100.0%	1.39	[1.31, 1.47]	0.36	0.60
HE Qw1.5B	100.0%	1.10	[1.07, 1.13]	0.84	0.91
TrQA Qw3B	100.0%	0.72	[0.68, 0.75]	0.46	0.68
TrQA Qw1.5B	100.0%	0.75	[0.72, 0.78]	0.59	0.68
TrQA Phi-3	100.0%	0.70	[0.65, 0.75]	0.29	0.49
TrQA TinyLL	100.0%	0.03	[0.02, 0.03]	0.03	0.15
i.i.d. uniform weights (simulation)	100%	1.31	—	—	—
random causal attention, no sink (simulation)	100%	1.27	—	—	—

Sequence lengths. Table 8 gives class-conditional lengths. TruthfulQA is length-balanced by construction; HaluEval’s hallucinated answers are systematically longer; on-policy, incorrect answers differ in length from correct ones because they are the model’s own.

Table 8: Class-conditional sequence length (mean $\pm$ SD, full sequence) and mean answer length.

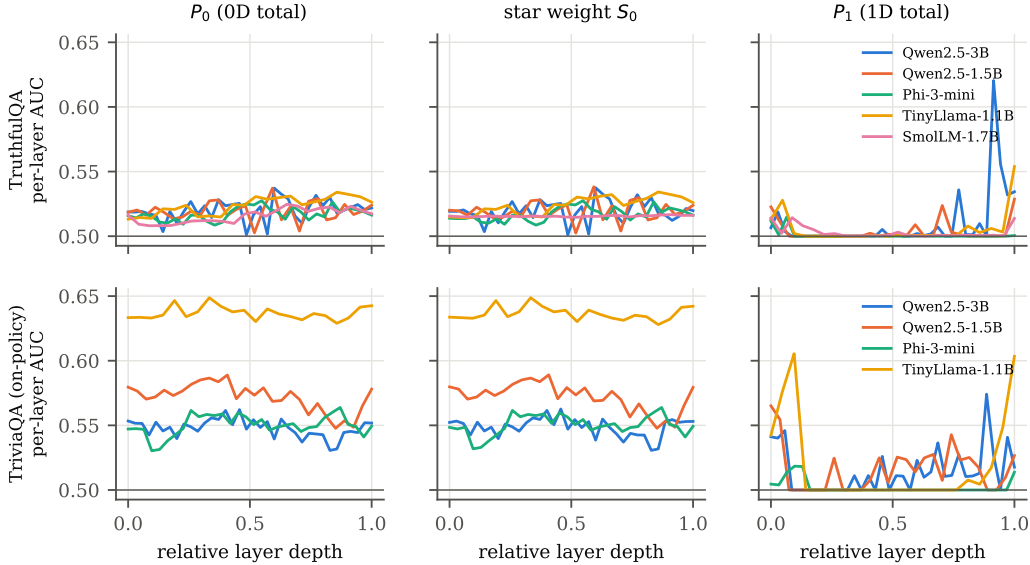
Setting	N (correct)	N (hallucinated)	answer tokens (c./h.)	point-biserial r	Welch p
TQA Qw3B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Qw1.5B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Phi-3	29.2 $\pm$ 8.9	28.7 $\pm$ 8.8	11.8 / 11.4	-0.024	0.32
TQA TinyLL	42.5 $\pm$ 9.0	42.1 $\pm$ 8.9	12.1 / 11.7	-0.024	0.32
TQA SmolLM	32.6 $\pm$ 8.2	32.1 $\pm$ 8.2	10.6 / 10.2	-0.026	0.29
HE Qw3B	121.2 $\pm$ 39.7	132.7 $\pm$ 41.2	3.8 / 15.4	+0.141	$< 10^{-9}$
HE Qw1.5B	122.2 $\pm$ 39.5	133.9 $\pm$ 41.3	3.6 / 15.3	+0.144	$< 10^{-5}$
TrQA Qw3B	68.0 $\pm$ 10.4	69.0 $\pm$ 9.7	5.3 / 5.7	+0.052	0.022
TrQA Qw1.5B	67.5 $\pm$ 10.3	69.4 $\pm$ 10.0	5.4 / 5.9	+0.086	0.00013
TrQA Phi-3	45.6 $\pm$ 11.3	46.9 $\pm$ 11.2	5.6 / 6.5	+0.057	0.01
TrQA TinyLL	67.3 $\pm$ 12.5	73.7 $\pm$ 14.9	17.0 / 19.6	+0.196	$< 10^{-20}$

Margin sensitivity. Table 9 reports TOST p -values at four equivalence margins for the four comparisons of Figure 5; bold marks $p < 0.05$ (equivalence).

Table 9: **TOST p -values by margin m .** NT: all non-topological features.

Setting m	Sink vs. 0D					0D Sink					1D 0D					0D NT				
	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025
TQA Qw3B	.029	.001	<.001	<.001	.040	.001	<.001	<.001	1.000	.995	.972	.882	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA Qw1.5B	.109	.016	<.001	<.001	.692	.276	.040	.002	.311	.034	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA Phi-3	.138	.013	<.001	<.001	.112	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA TinyLL	.520	.234	.064	.012	.962	.809	.483	.169	.210	.021	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA SmolLM	.939	.865	.748	.602	1.000	.996	.960	.805	.179	.036	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
HE Qw3B	.910	.676	.351	.110	1.000	1.000	.982	.804	.906	.551	.148	.012	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
HE Qw1.5B	.993	.970	.896	.755	.679	.263	.041	.002	.980	.918	.786	.569	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Qw3B	.792	.522	.239	.069	.982	.866	.561	.219	.106	.015	.002	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Qw1.5B	.431	.176	.050	.010	.528	.211	.044	.005	.079	.011	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Phi-3	.047	.002	<.001	<.001	.114	.005	<.001	<.001	.086	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TrQA TinyLL	.091	.021	.003	<.001	.071	.005	<.001	<.001	.083	.013	.002	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001

Per-layer signal. Figure 2 shows single-feature AUCs by depth. P_0 and the star weight trace nearly identical curves in every model with a sink, while P_1 carries signal only in the layers that are not coned. These AUCs are orientation-free ($\max(\text{AUC}, 1 - \text{AUC})$) and therefore slightly optimistic; they are descriptive, not a detector.

Figure 2: **Single-feature AUC by layer.**

Worked example. Figure 3 shows TruthfulQA question 67 with its incorrect answer (“Yes, vampires are real”) under TinyLlama-1.1B ($N=29$). In layer 5 the model sends 0.85 of its attention to token 0 on average, $\delta_0 = 0.000$, and accordingly $P_0 = 4.225$ equals the star weight $S_0 = 4.225$ and there are 0 H_1 bars. In layer 21 sink attention drops to 0.53 and $\delta_0 = 0.281$; now $P_0 = 12.287$ falls below $S_0 = 13.043$ (within the bound $(N-1)\delta_0 = 7.87$), and 10 H_1 bars appear, the longest of length $0.172 \leq \delta_0$. The theorem bounds loops but does not force them: in layers 2 and 11–18 of the same example δ_0 lies between 0.11 and 0.52, yet H_1 is empty.

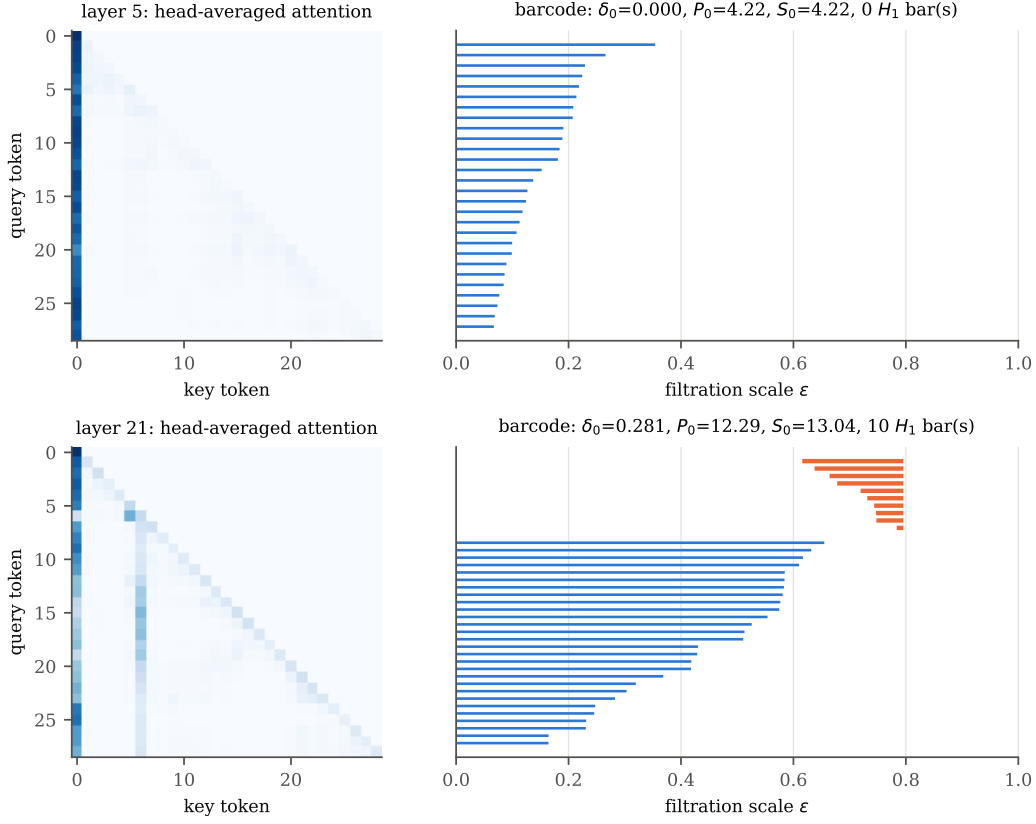


Figure 3: **A coned and a non-coned layer of the same example** (TinyLlama-1.1B, TruthfulQA). Left: head-averaged attention; the bright first column is the sink. Right: 0D bars (blue) and 1D bars (orange) of the Vietoris-Rips filtration of $D = 1 - \max(A, A^T)$.

D ONE-DIMENSIONAL PERSISTENCE

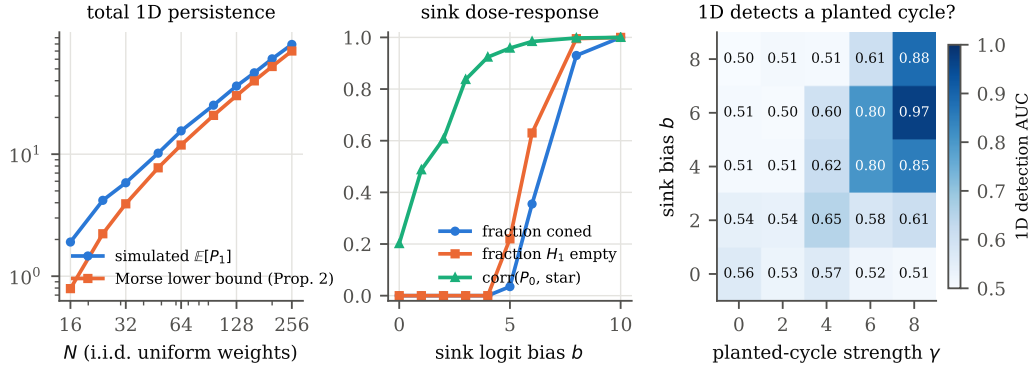


Figure 4: **Synthetic checks.** Left: Proposition 2’s lower bound against the simulated mean total 1D persistence under i.i.d. Uniform $[0, 1]$ weights (20 draws per $N \leq 128$, 8 above). Middle: random causal attention ($N=40$, 200 draws per point) with a logit bias b on token 0: as the sink strengthens, graphs become exactly coned, H_1 empties, and P_0 becomes a function of the star weight. Right: AUC of 1D total persistence for detecting a planted 6-cycle of logit strength γ (150 draws per class). Without a sink ($b=0$) the planted cycle is lost among background cycles; under a strong sink ($b=8$) it is invisible unless it is strong enough to break the cone.

Under i.i.d. weights, 1D persistence grows faster than linearly over the lengths that matter here and linearly asymptotically (Proposition 2; Figure 4, left). Real attention behaves very differently: most graphs have *no* 1D bars at all (Table 1), and the bars that exist are confined to graphs with a non-zero coning defect. Figure 4 (middle) reproduces this transition in a controlled setting: raising the sink’s logit bias moves random causal attention from the i.i.d.-like regime to exact coning, with H_1 emptying in lockstep.

This explains a recurring empirical observation, that 1D features add little to 0D. At full sample size, adding length-normalized 1D to 0D changes AUC by -0.008 to $+0.033$ and is equivalent within ± 0.015 in 9 of 13 sink settings; the exceptions are TQA Qw3B, HE Qw3B, HE Qw1.5B and TrQA Mis7B, where adding 1D helps significantly. The TruthfulQA Qwen2.5-3B gain is also the result that is most sensitive to inference precision (Appendix B). The mechanism is not that loops are uninformative in principle. In a strongly sinked model, a loop can only exist where some token pays another token more attention than one of the two pays to the sink, and its lifetime is bounded by that excess. Figure 4 (right) makes this concrete. A planted 6-cycle that H_1 detects easily under a moderate sink ($b=6$, $\gamma=8$: AUC 0.97) is invisible under a strong one ($b=8$, $\gamma \leq 4$: AUC ≤ 0.51) until it is strong enough to break the cone. Without any sink ($b=0$) H_1 also misses it (AUC ≤ 0.57), because random attention already has many background cycles; 1D persistence is informative only in between.

E FULL RESULTS

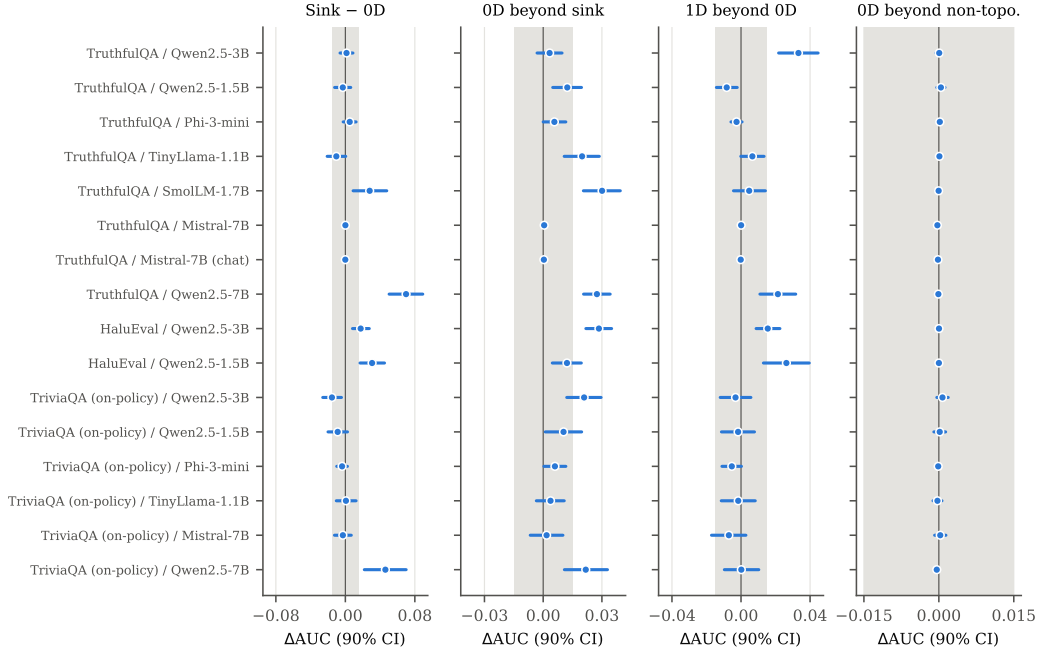


Figure 5: **Four comparisons across all settings** (Δ AUC with 90% cluster-bootstrap intervals; shaded band: ± 0.015). Sink features nearly reproduce 0D; 0D adds little beyond them; 1D adds little beyond 0D except in three settings; and 0D adds nothing beyond non-topological probes under early fusion (late fusion in Table 10).

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	LEN	2	0.529	0.0003	0.567
TQA Qw3B	LEX	—	0.820	0.0014	0.829
TQA Qw3B	LOGPROB	6	0.630	0.0006	0.694
TQA Qw3B	LLMCHECK	36	0.693	0.0011	0.764
TQA Qw3B	ROWSTAT	72	0.750	0.0003	0.823
TQA Qw3B	SINK	72	0.737	0.0022	0.767

	Setting	Bank	dim	AUC	seed SD	pair acc.
1134						
1135						
1136	TQA Qw3B	0D	72	0.736	0.0013	0.778
1137	TQA Qw3B	LEGACY MSTPROXY	36	0.713	0.0008	0.788
1138	TQA Qw3B	1D	144	0.660	0.0001	0.674
1139	TQA Qw3B	DEFL	216	0.715	0.0028	0.753
1140	TQA Qw3B	ANS	216	0.748	0.0017	0.749
1141	TQA Qw3B	PH SINK	576	0.851	0.0015	0.870
1142	TQA Qw3B	PH 0DTOT	576	0.842	0.0011	0.869
1143	TQA Qw3B	PH 0D	1152	0.853	0.0013	0.862
1144	TQA Qw3B	PH ENT	576	0.857	0.0013	0.901
1145	TQA Qw3B	LOOKBACK	576	0.897	0.0022	0.908
1146	TQA Qw3B	HIDDEN	2048	0.913	0.0008	0.918
1147	TQA Qw3B	NONTOPO	2738	0.926	0.0006	0.936
1148	TQA Qw3B	0D+1D	216	0.769	0.0004	0.791
1149	TQA Qw3B	0D+SINK	144	0.741	0.0015	0.776
1150	TQA Qw3B	HIDDEN+0D	2120	0.913	0.0007	0.920
1151	TQA Qw3B	LEX+0D	–	0.840	0.0015	0.848
1152	TQA Qw3B	LEX+DEFL	–	0.830	0.0004	0.854
1153	TQA Qw3B	LEX+SINK	–	0.843	0.0020	0.851
1154	TQA Qw3B	LOGPROB+0D	78	0.776	0.0022	0.810
1155	TQA Qw3B	NONTOPO+0D	2810	0.926	0.0004	0.936
1156	TQA Qw3B	NONTOPO+DEFL	2954	0.928	0.0006	0.933
1157	TQA Qw3B	PH SINK+PH 0DTOT	1152	0.860	0.0022	0.892
1158	TQA Qw3B	SINK+1D	216	0.766	0.0007	0.787
1159	TQA Qw3B	SINK+ANS	288	0.785	0.0031	0.797
1160	TQA Qw3B	SINK+DEFL	288	0.766	0.0024	0.799
1161	TQA Qw1.5B	LEN	2	0.529	0.0003	0.567
1162	TQA Qw1.5B	LEX	–	0.820	0.0014	0.829
1163	TQA Qw1.5B	LOGPROB	6	0.638	0.0007	0.682
1164	TQA Qw1.5B	LLMCHECK	28	0.666	0.0005	0.743
1165	TQA Qw1.5B	ROWSTAT	56	0.738	0.0014	0.815
1166	TQA Qw1.5B	SINK	56	0.723	0.0019	0.771
1167	TQA Qw1.5B	0D	56	0.726	0.0007	0.767
1168	TQA Qw1.5B	LEGACY MSTPROXY	28	0.698	0.0010	0.786
1169	TQA Qw1.5B	1D	112	0.586	0.0014	0.644
1170	TQA Qw1.5B	DEFL	168	0.703	0.0042	0.755
1171	TQA Qw1.5B	ANS	168	0.697	0.0040	0.700
1172	TQA Qw1.5B	PH SINK	336	0.830	0.0008	0.851
1173	TQA Qw1.5B	PH 0DTOT	336	0.819	0.0011	0.838
1174	TQA Qw1.5B	PH 0D	672	0.838	0.0012	0.847
1175	TQA Qw1.5B	PH ENT	336	0.833	0.0025	0.852
1176	TQA Qw1.5B	LOOKBACK	336	0.851	0.0020	0.856
1177	TQA Qw1.5B	HIDDEN	1536	0.916	0.0012	0.923
1178	TQA Qw1.5B	NONTOPO	1962	0.917	0.0013	0.928
1179	TQA Qw1.5B	0D+1D	168	0.718	0.0013	0.767
1180	TQA Qw1.5B	0D+SINK	112	0.735	0.0012	0.766
1181	TQA Qw1.5B	HIDDEN+0D	1592	0.916	0.0014	0.924
1182	TQA Qw1.5B	LEX+0D	–	0.835	0.0017	0.841
1183	TQA Qw1.5B	LEX+DEFL	–	0.825	0.0019	0.836
1184	TQA Qw1.5B	LEX+SINK	–	0.833	0.0008	0.846
1185	TQA Qw1.5B	LOGPROB+0D	62	0.740	0.0013	0.797
1186	TQA Qw1.5B	NONTOPO+0D	2018	0.917	0.0014	0.927
1187	TQA Qw1.5B	NONTOPO+DEFL	2130	0.917	0.0014	0.923
	TQA Qw1.5B	PH SINK+PH 0DTOT	672	0.841	0.0013	0.867
	TQA Qw1.5B	SINK+1D	168	0.715	0.0017	0.769
	TQA Qw1.5B	SINK+ANS	224	0.761	0.0027	0.753
	TQA Qw1.5B	SINK+DEFL	224	0.757	0.0028	0.802
	TQA Phi-3	LEN	2	0.522	0.0004	0.562
	TQA Phi-3	LEX	–	0.820	0.0014	0.829
	TQA Phi-3	LOGPROB	6	0.578	0.0016	0.597
	TQA Phi-3	LLMCHECK	32	0.653	0.0008	0.755
	TQA Phi-3	ROWSTAT	64	0.661	0.0011	0.738
	TQA Phi-3	SINK	64	0.651	0.0006	0.700
	TQA Phi-3	0D	64	0.646	0.0011	0.709

	Setting	Bank	dim	AUC	seed SD	pair acc.
1188						
1189						
1190	TQA Phi-3	LEGACY MSTPROXY	32	0.613	0.0008	0.694
1191	TQA Phi-3	1D	128	0.526	0.0030	0.551
1192	TQA Phi-3	DEFL	192	0.639	0.0011	0.684
1193	TQA Phi-3	ANS	192	0.629	0.0042	0.651
1194	TQA Phi-3	PH SINK	1024	0.871	0.0008	0.901
1195	TQA Phi-3	PH 0DTOT	1024	0.845	0.0014	0.882
1196	TQA Phi-3	PH 0D	2048	0.868	0.0007	0.881
1197	TQA Phi-3	PH ENT	1024	0.864	0.0010	0.895
1198	TQA Phi-3	LOOKBACK	1024	0.892	0.0004	0.902
1199	TQA Phi-3	HIDDEN	3072	0.933	0.0007	0.945
1200	TQA Phi-3	NONTOPO	4198	0.936	0.0006	0.942
1201	TQA Phi-3	0D+1D	192	0.643	0.0014	0.705
1202	TQA Phi-3	0D+SINK	128	0.657	0.0007	0.704
1203	TQA Phi-3	HIDDEN+0D	3136	0.933	0.0008	0.944
1204	TQA Phi-3	LEX+0D	–	0.821	0.0003	0.842
1205	TQA Phi-3	LEX+DEFL	–	0.802	0.0001	0.827
1206	TQA Phi-3	LEX+SINK	–	0.820	0.0007	0.834
1207	TQA Phi-3	LOGPROB+0D	70	0.655	0.0011	0.694
1208	TQA Phi-3	NONTOPO+0D	4262	0.936	0.0006	0.942
1209	TQA Phi-3	NONTOPO+DEFL	4390	0.937	0.0007	0.939
1210	TQA Phi-3	PH SINK+PH 0DTOT	2048	0.871	0.0007	0.905
1211	TQA Phi-3	SINK+1D	192	0.651	0.0011	0.696
1212	TQA Phi-3	SINK+ANS	256	0.675	0.0015	0.698
1213	TQA Phi-3	SINK+DEFL	256	0.684	0.0023	0.718
1214	TQA TinyLl.	LEN	2	0.523	0.0005	0.556
1215	TQA TinyLl.	LEX	–	0.820	0.0014	0.829
1216	TQA TinyLl.	LOGPROB	6	0.637	0.0010	0.645
1217	TQA TinyLl.	LLMCHECK	22	0.614	0.0005	0.705
1218	TQA TinyLl.	ROWSTAT	44	0.661	0.0015	0.749
1219	TQA TinyLl.	SINK	44	0.643	0.0006	0.718
1220	TQA TinyLl.	0D	44	0.654	0.0011	0.729
1221	TQA TinyLl.	LEGACY MSTPROXY	22	0.627	0.0009	0.741
1222	TQA TinyLl.	1D	88	0.592	0.0006	0.673
1223	TQA TinyLl.	DEFL	132	0.651	0.0019	0.705
1224	TQA TinyLl.	ANS	132	0.644	0.0049	0.678
1225	TQA TinyLl.	PH SINK	704	0.808	0.0015	0.841
1226	TQA TinyLl.	PH 0DTOT	704	0.804	0.0021	0.830
1227	TQA TinyLl.	PH 0D	1408	0.806	0.0015	0.820
1228	TQA TinyLl.	PH ENT	704	0.817	0.0008	0.838
1229	TQA TinyLl.	LOOKBACK	704	0.850	0.0028	0.846
1230	TQA TinyLl.	HIDDEN	2048	0.883	0.0013	0.879
1231	TQA TinyLl.	NONTOPO	2824	0.885	0.0011	0.887
1232	TQA TinyLl.	0D+1D	132	0.660	0.0010	0.744
1233	TQA TinyLl.	0D+SINK	88	0.663	0.0004	0.728
1234	TQA TinyLl.	HIDDEN+0D	2092	0.883	0.0014	0.880
1235	TQA TinyLl.	LEX+0D	–	0.830	0.0012	0.836
1236	TQA TinyLl.	LEX+DEFL	–	0.826	0.0012	0.846
1237	TQA TinyLl.	LEX+SINK	–	0.831	0.0011	0.832
1238	TQA TinyLl.	LOGPROB+0D	50	0.683	0.0010	0.711
1239	TQA TinyLl.	NONTOPO+0D	2868	0.885	0.0011	0.889
1240	TQA TinyLl.	NONTOPO+DEFL	2956	0.884	0.0010	0.885
1241	TQA TinyLl.	PH SINK+PH 0DTOT	1408	0.810	0.0031	0.841
	TQA TinyLl.	SINK+1D	132	0.651	0.0016	0.729
	TQA TinyLl.	SINK+ANS	176	0.685	0.0041	0.720
	TQA TinyLl.	SINK+DEFL	176	0.688	0.0015	0.749
	TQA SmolLM	LEN	2	0.524	0.0003	0.554
	TQA SmolLM	LEX	–	0.820	0.0014	0.829
	TQA SmolLM	LOGPROB	6	0.588	0.0010	0.592
	TQA SmolLM	LLMCHECK	24	0.604	0.0004	0.684
	TQA SmolLM	ROWSTAT	48	0.673	0.0014	0.741
	TQA SmolLM	SINK	48	0.611	0.0019	0.628
	TQA SmolLM	0D	48	0.583	0.0029	0.656
	TQA SmolLM	LEGACY MSTPROXY	24	0.585	0.0011	0.674

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA SmolLM	1D	96	0.543	0.0015	0.581
TQA SmolLM	DEFL	144	0.594	0.0023	0.650
TQA SmolLM	ANS	144	0.628	0.0041	0.630
TQA SmolLM	PH SINK	768	0.720	0.0008	0.775
TQA SmolLM	PH 0DTOT	768	0.819	0.0019	0.845
TQA SmolLM	PH 0D	1536	0.800	0.0012	0.818
TQA SmolLM	PH ENT	768	0.834	0.0007	0.841
TQA SmolLM	LOOKBACK	768	0.863	0.0017	0.864
TQA SmolLM	HIDDEN	2048	0.894	0.0022	0.892
TQA SmolLM	NONTOPO	2894	0.899	0.0015	0.903
TQA SmolLM	0D+1D	144	0.588	0.0019	0.643
TQA SmolLM	0D+SINK	96	0.641	0.0026	0.662
TQA SmolLM	HIDDEN+0D	2096	0.894	0.0020	0.891
TQA SmolLM	LEX+0D	–	0.827	0.0008	0.838
TQA SmolLM	LEX+DEFL	–	0.822	0.0001	0.845
TQA SmolLM	LEX+SINK	–	0.819	0.0015	0.826
TQA SmolLM	LOGPROB+0D	54	0.613	0.0009	0.630
TQA SmolLM	NONTOPO+0D	2942	0.899	0.0013	0.902
TQA SmolLM	NONTOPO+DEFL	3038	0.899	0.0012	0.900
TQA SmolLM	PH SINK+PH 0DTOT	1536	0.826	0.0023	0.854
TQA SmolLM	SINK+1D	144	0.629	0.0013	0.650
TQA SmolLM	SINK+ANS	192	0.646	0.0021	0.651
TQA SmolLM	SINK+DEFL	192	0.652	0.0022	0.681
TQA Mis7B	LEN	2	0.525	0.0002	0.571
TQA Mis7B	LEX	–	0.820	0.0008	0.828
TQA Mis7B	LOGPROB	6	0.641	0.0004	0.652
TQA Mis7B	LLMCHECK	32	0.652	0.0002	0.770
TQA Mis7B	ROWSTAT	64	0.694	0.0009	0.821
TQA Mis7B	SINK	64	0.685	0.0008	0.755
TQA Mis7B	0D	64	0.684	0.0008	0.755
TQA Mis7B	LEGACY MSTPROXY	32	0.646	0.0009	0.769
TQA Mis7B	1D	128	0.501	0.0000	0.499
TQA Mis7B	DEFL	192	0.710	0.0015	0.756
TQA Mis7B	ANS	192	0.677	0.0043	0.704
TQA Mis7B	PH SINK	1024	0.903	0.0009	0.938
TQA Mis7B	PH 0DTOT	1024	0.887	0.0020	0.928
TQA Mis7B	PH 0D	2048	0.893	0.0013	0.908
TQA Mis7B	PH ENT	1024	0.897	0.0010	0.929
TQA Mis7B	LOOKBACK	1024	0.933	0.0008	0.945
TQA Mis7B	HIDDEN	4096	0.956	0.0005	0.957
TQA Mis7B	NONTOPO	5222	0.959	0.0006	0.967
TQA Mis7B	0D+1D	192	0.685	0.0007	0.760
TQA Mis7B	0D+SINK	128	0.685	0.0007	0.754
TQA Mis7B	HIDDEN+0D	4160	0.956	0.0005	0.956
TQA Mis7B	LEX+0D	–	0.834	0.0013	0.859
TQA Mis7B	LEX+DEFL	–	0.823	0.0013	0.856
TQA Mis7B	LEX+SINK	–	0.833	0.0015	0.859
TQA Mis7B	LOGPROB+0D	70	0.722	0.0012	0.775
TQA Mis7B	NONTOPO+0D	5286	0.958	0.0006	0.967
TQA Mis7B	NONTOPO+DEFL	5414	0.959	0.0006	0.966
TQA Mis7B	PH SINK+PH 0DTOT	2048	0.903	0.0013	0.936
TQA Mis7B	SINK+1D	192	0.685	0.0008	0.758
TQA Mis7B	SINK+ANS	256	0.727	0.0023	0.763
TQA Mis7B	SINK+DEFL	256	0.740	0.0004	0.800
TQA Mis7B (chat)	LEN	2	0.525	0.0002	0.571
TQA Mis7B (chat)	LEX	–	0.820	0.0008	0.828
TQA Mis7B (chat)	LOGPROB	6	0.643	0.0010	0.646
TQA Mis7B (chat)	LLMCHECK	32	0.653	0.0001	0.769
TQA Mis7B (chat)	ROWSTAT	64	0.700	0.0008	0.823
TQA Mis7B (chat)	SINK	64	0.688	0.0010	0.754
TQA Mis7B (chat)	0D	64	0.688	0.0010	0.755
TQA Mis7B (chat)	LEGACY MSTPROXY	32	0.652	0.0009	0.783
TQA Mis7B (chat)	1D	128	0.501	0.0000	0.499

	Setting	Bank	dim	AUC	seed SD	pair acc.
1296						
1297						
1298	TQA Mis7B (chat)	DEFL	192	0.733	0.0013	0.781
1299	TQA Mis7B (chat)	ANS	192	0.723	0.0033	0.736
1300	TQA Mis7B (chat)	PH SINK	1024	0.907	0.0011	0.934
1301	TQA Mis7B (chat)	PH 0DTOT	1024	0.890	0.0021	0.931
1302	TQA Mis7B (chat)	PH 0D	2048	0.900	0.0017	0.919
1303	TQA Mis7B (chat)	PH ENT	1024	0.901	0.0010	0.928
1304	TQA Mis7B (chat)	LOOKBACK	1024	0.930	0.0008	0.942
1305	TQA Mis7B (chat)	HIDDEN	4096	0.958	0.0004	0.956
1306	TQA Mis7B (chat)	NONTOPO	5222	0.960	0.0003	0.957
1307	TQA Mis7B (chat)	0D+1D	192	0.688	0.0009	0.754
1308	TQA Mis7B (chat)	0D+SINK	128	0.689	0.0008	0.755
1309	TQA Mis7B (chat)	HIDDEN+0D	4160	0.959	0.0004	0.957
1310	TQA Mis7B (chat)	LEX+0D	–	0.834	0.0013	0.860
1311	TQA Mis7B (chat)	LEX+DEFL	–	0.835	0.0020	0.864
1312	TQA Mis7B (chat)	LEX+SINK	–	0.834	0.0015	0.859
1313	TQA Mis7B (chat)	LOGPROB+0D	70	0.729	0.0021	0.793
1314	TQA Mis7B (chat)	NONTOPO+0D	5286	0.960	0.0003	0.958
1315	TQA Mis7B (chat)	NONTOPO+DEFL	5414	0.961	0.0005	0.960
1316	TQA Mis7B (chat)	PH SINK+PH 0DTOT	2048	0.907	0.0012	0.938
1317	TQA Mis7B (chat)	SINK+1D	192	0.688	0.0009	0.756
1318	TQA Mis7B (chat)	SINK+ANS	256	0.751	0.0035	0.788
1319	TQA Mis7B (chat)	SINK+DEFL	256	0.756	0.0018	0.803
1320	TQA Qw7B	LEN	2	0.529	0.0003	0.567
1321	TQA Qw7B	LEX	–	0.820	0.0008	0.828
1322	TQA Qw7B	LOGPROB	6	0.674	0.0007	0.717
1323	TQA Qw7B	LLMCHECK	28	0.667	0.0011	0.729
1324	TQA Qw7B	ROWSTAT	56	0.736	0.0008	0.807
1325	TQA Qw7B	SINK	56	0.749	0.0001	0.761
1326	TQA Qw7B	0D	56	0.679	0.0010	0.774
1327	TQA Qw7B	LEGACY MSTPROXY	28	0.678	0.0011	0.785
1328	TQA Qw7B	1D	112	0.610	0.0005	0.634
1329	TQA Qw7B	DEFL	168	0.697	0.0009	0.748
1330	TQA Qw7B	ANS	168	0.741	0.0019	0.749
1331	TQA Qw7B	PH SINK	784	0.821	0.0025	0.843
1332	TQA Qw7B	PH 0DTOT	784	0.872	0.0016	0.909
1333	TQA Qw7B	PH 0D	1568	0.850	0.0022	0.867
1334	TQA Qw7B	PH ENT	784	0.887	0.0003	0.901
1335	TQA Qw7B	LOOKBACK	784	0.922	0.0004	0.923
1336	TQA Qw7B	HIDDEN	3584	0.933	0.0008	0.934
1337	TQA Qw7B	NONTOPO	4458	0.950	0.0013	0.950
1338	TQA Qw7B	0D+1D	168	0.701	0.0002	0.756
1339	TQA Qw7B	0D+SINK	112	0.777	0.0004	0.794
1340	TQA Qw7B	HIDDEN+0D	3640	0.933	0.0008	0.933
1341	TQA Qw7B	LEX+0D	–	0.834	0.0008	0.858
1342	TQA Qw7B	LEX+DEFL	–	0.828	0.0014	0.844
1343	TQA Qw7B	LEX+SINK	–	0.848	0.0001	0.849
1344	TQA Qw7B	LOGPROB+0D	62	0.727	0.0002	0.792
1345	TQA Qw7B	NONTOPO+0D	4514	0.949	0.0013	0.949
1346	TQA Qw7B	NONTOPO+DEFL	4626	0.949	0.0012	0.952
1347	TQA Qw7B	PH SINK+PH 0DTOT	1568	0.881	0.0005	0.920
1348	TQA Qw7B	SINK+1D	168	0.767	0.0023	0.780
1349	TQA Qw7B	SINK+ANS	224	0.794	0.0010	0.802
	TQA Qw7B	SINK+DEFL	224	0.788	0.0006	0.797
	HE Qw3B	LEN	2	0.962	0.0001	0.970
	HE Qw3B	LEX	–	0.981	0.0004	0.977
	HE Qw3B	LOGPROB	6	0.993	0.0001	0.991
	HE Qw3B	LLMCHECK	36	0.745	0.0004	0.956
	HE Qw3B	ROWSTAT	72	0.829	0.0002	0.976
	HE Qw3B	SINK	72	0.829	0.0008	0.930
	HE Qw3B	0D	72	0.811	0.0006	0.927
	HE Qw3B	LEGACY MSTPROXY	36	0.731	0.0002	0.960
	HE Qw3B	1D	144	0.710	0.0013	0.891
	HE Qw3B	DEFL	216	0.834	0.0009	0.959

	Setting	Bank	dim	AUC	seed SD	pair acc.
1350						
1351						
1352	HE Qw3B	ANS	216	0.978	0.0009	0.980
1353	HE Qw3B	PH SINK	576	0.958	0.0007	0.999
1354	HE Qw3B	PH 0DTOT	576	0.958	0.0005	0.995
1355	HE Qw3B	PH 0D	1152	0.971	0.0006	0.988
1356	HE Qw3B	PH ENT	576	0.964	0.0005	0.993
1357	HE Qw3B	LOOKBACK	576	0.998	0.0001	0.998
1358	HE Qw3B	HIDDEN	2048	1.000	0.0000	1.000
1359	HE Qw3B	NONTOPO	2738	1.000	0.0002	1.000
1360	HE Qw3B	0D+1D	216	0.826	0.0013	0.945
1361	HE Qw3B	0D+SINK	144	0.857	0.0010	0.948
1362	HE Qw3B	HIDDEN+0D	2120	1.000	0.0000	1.000
1363	HE Qw3B	LEX+0D	–	0.979	0.0001	0.983
1364	HE Qw3B	LEX+DEFL	–	0.971	0.0003	0.975
1365	HE Qw3B	LEX+SINK	–	0.980	0.0003	0.981
1366	HE Qw3B	LOGPROB+0D	78	0.993	0.0003	0.992
1367	HE Qw3B	NONTOPO+0D	2810	1.000	0.0002	1.000
1368	HE Qw3B	NONTOPO+DEFL	2954	1.000	0.0002	1.000
1369	HE Qw3B	PH SINK+PH 0DTOT	1152	0.962	0.0006	0.997
1370	HE Qw3B	SINK+1D	216	0.843	0.0011	0.938
1371	HE Qw3B	SINK+ANS	288	0.978	0.0003	0.982
1372	HE Qw3B	SINK+DEFL	288	0.887	0.0020	0.957
1373	HE Qw1.5B	LEN	2	0.967	0.0002	0.976
1374	HE Qw1.5B	LEX	–	0.979	0.0004	0.980
1375	HE Qw1.5B	LOGPROB	6	0.995	0.0002	0.988
1376	HE Qw1.5B	LLMCHECK	28	0.791	0.0005	0.978
1377	HE Qw1.5B	ROWSTAT	56	0.788	0.0017	0.984
1378	HE Qw1.5B	SINK	56	0.820	0.0015	0.934
1379	HE Qw1.5B	0D	56	0.789	0.0014	0.918
1380	HE Qw1.5B	LEGACY MSTPROXY	28	0.719	0.0003	0.976
1381	HE Qw1.5B	1D	112	0.747	0.0006	0.878
1382	HE Qw1.5B	DEFL	168	0.825	0.0021	0.952
1383	HE Qw1.5B	ANS	168	0.972	0.0006	0.978
1384	HE Qw1.5B	PH SINK	336	0.922	0.0007	0.998
1385	HE Qw1.5B	PH 0DTOT	336	0.921	0.0007	0.996
1386	HE Qw1.5B	PH 0D	672	0.949	0.0027	0.974
1387	HE Qw1.5B	PH ENT	336	0.933	0.0011	0.996
1388	HE Qw1.5B	LOOKBACK	336	0.999	0.0002	0.998
1389	HE Qw1.5B	HIDDEN	1536	1.000	0.0001	1.000
1390	HE Qw1.5B	NONTOPO	1962	1.000	0.0002	1.000
1391	HE Qw1.5B	0D+1D	168	0.815	0.0032	0.922
1392	HE Qw1.5B	0D+SINK	112	0.832	0.0015	0.924
1393	HE Qw1.5B	HIDDEN+0D	1592	1.000	0.0001	1.000
1394	HE Qw1.5B	LEX+0D	–	0.971	0.0004	0.986
1395	HE Qw1.5B	LEX+DEFL	–	0.956	0.0002	0.986
1396	HE Qw1.5B	LEX+SINK	–	0.968	0.0005	0.984
1397	HE Qw1.5B	LOGPROB+0D	62	0.994	0.0002	0.990
1398	HE Qw1.5B	NONTOPO+0D	2018	1.000	0.0000	1.000
1399	HE Qw1.5B	NONTOPO+DEFL	2130	1.000	0.0001	1.000
1400	HE Qw1.5B	PH SINK+PH 0DTOT	672	0.931	0.0024	0.998
1401	HE Qw1.5B	SINK+1D	168	0.826	0.0035	0.940
1402	HE Qw1.5B	SINK+ANS	224	0.975	0.0015	0.990
1403	HE Qw1.5B	SINK+DEFL	224	0.872	0.0028	0.962
	TrQA Qw3B	LEN	2	0.569	0.0014	–
	TrQA Qw3B	LEX	–	0.596	0.0039	–
	TrQA Qw3B	LOGPROB	6	0.839	0.0002	–
	TrQA Qw3B	LLMCHECK	36	0.688	0.0012	–
	TrQA Qw3B	ROWSTAT	72	0.724	0.0046	–
	TrQA Qw3B	SINK	72	0.717	0.0038	–
	TrQA Qw3B	0D	72	0.732	0.0038	–
	TrQA Qw3B	LEGACY MSTPROXY	36	0.691	0.0033	–
	TrQA Qw3B	1D	144	0.601	0.0031	–
	TrQA Qw3B	DEFL	216	0.684	0.0043	–
	TrQA Qw3B	ANS	216	0.719	0.0020	–

	Setting	Bank	dim	AUC	seed SD	pair acc.
1404						
1405						
1406	TrQA Qw3B	PH SINK	576	0.820	0.0021	—
1407	TrQA Qw3B	PH 0DTOT	576	0.807	0.0012	—
1408	TrQA Qw3B	PH 0D	1152	0.807	0.0017	—
1409	TrQA Qw3B	PH ENT	576	0.810	0.0007	—
1410	TrQA Qw3B	LOOKBACK	576	0.861	0.0013	—
1411	TrQA Qw3B	HIDDEN	2048	0.841	0.0042	—
1412	TrQA Qw3B	NONTOPO	2738	0.870	0.0028	—
1413	TrQA Qw3B	0D+1D	216	0.729	0.0022	—
1414	TrQA Qw3B	0D+SINK	144	0.738	0.0050	—
1415	TrQA Qw3B	HIDDEN+0D	2120	0.843	0.0040	—
1416	TrQA Qw3B	LEX+0D	—	0.733	0.0011	—
1417	TrQA Qw3B	LEX+DEFL	—	0.690	0.0031	—
1418	TrQA Qw3B	LEX+SINK	—	0.724	0.0005	—
1419	TrQA Qw3B	LOGPROB+0D	78	0.843	0.0022	—
1420	TrQA Qw3B	NONTOPO+0D	2810	0.870	0.0027	—
1421	TrQA Qw3B	NONTOPO+DEFL	2954	0.870	0.0029	—
1422	TrQA Qw3B	PH SINK+PH 0DTOT	1152	0.824	0.0017	—
1423	TrQA Qw3B	SINK+1D	216	0.718	0.0035	—
1424	TrQA Qw3B	SINK+ANS	288	0.752	0.0010	—
1425	TrQA Qw3B	SINK+DEFL	288	0.719	0.0038	—
1426	TrQA Qw1.5B	LEN	2	0.588	0.0012	—
1427	TrQA Qw1.5B	LEX	—	0.609	0.0033	—
1428	TrQA Qw1.5B	LOGPROB	6	0.823	0.0006	—
1429	TrQA Qw1.5B	LLMCHECK	28	0.676	0.0029	—
1430	TrQA Qw1.5B	ROWSTAT	56	0.720	0.0008	—
1431	TrQA Qw1.5B	SINK	56	0.707	0.0000	—
1432	TrQA Qw1.5B	0D	56	0.716	0.0004	—
1433	TrQA Qw1.5B	LEGACY MSTPROXY	28	0.685	0.0014	—
1434	TrQA Qw1.5B	1D	112	0.574	0.0024	—
1435	TrQA Qw1.5B	DEFL	168	0.655	0.0016	—
1436	TrQA Qw1.5B	ANS	168	0.675	0.0035	—
1437	TrQA Qw1.5B	PH SINK	336	0.767	0.0034	—
1438	TrQA Qw1.5B	PH 0DTOT	336	0.759	0.0051	—
1439	TrQA Qw1.5B	PH 0D	672	0.768	0.0013	—
1440	TrQA Qw1.5B	PH ENT	336	0.768	0.0025	—
1441	TrQA Qw1.5B	LOOKBACK	336	0.824	0.0017	—
1442	TrQA Qw1.5B	HIDDEN	1536	0.855	0.0031	—
1443	TrQA Qw1.5B	NONTOPO	1962	0.873	0.0023	—
1444	TrQA Qw1.5B	0D+1D	168	0.714	0.0015	—
1445	TrQA Qw1.5B	0D+SINK	112	0.717	0.0006	—
1446	TrQA Qw1.5B	HIDDEN+0D	1592	0.855	0.0032	—
1447	TrQA Qw1.5B	LEX+0D	—	0.721	0.0031	—
1448	TrQA Qw1.5B	LEX+DEFL	—	0.672	0.0047	—
1449	TrQA Qw1.5B	LEX+SINK	—	0.715	0.0036	—
1450	TrQA Qw1.5B	LOGPROB+0D	62	0.828	0.0007	—
1451	TrQA Qw1.5B	NONTOPO+0D	2018	0.873	0.0025	—
1452	TrQA Qw1.5B	NONTOPO+DEFL	2130	0.872	0.0024	—
1453	TrQA Qw1.5B	PH SINK+PH 0DTOT	672	0.772	0.0022	—
1454	TrQA Qw1.5B	SINK+1D	168	0.709	0.0028	—
1455	TrQA Qw1.5B	SINK+ANS	224	0.726	0.0059	—
1456	TrQA Qw1.5B	SINK+DEFL	224	0.719	0.0023	—
1457	TrQA Phi-3	LEN	2	0.581	0.0022	—
	TrQA Phi-3	LEX	—	0.635	0.0083	—
	TrQA Phi-3	LOGPROB	6	0.877	0.0002	—
	TrQA Phi-3	LLMCHECK	32	0.676	0.0018	—
	TrQA Phi-3	ROWSTAT	64	0.728	0.0006	—
	TrQA Phi-3	SINK	64	0.707	0.0018	—
	TrQA Phi-3	0D	64	0.710	0.0022	—
	TrQA Phi-3	LEGACY MSTPROXY	32	0.669	0.0002	—
	TrQA Phi-3	1D	128	0.546	0.0031	—
	TrQA Phi-3	DEFL	192	0.698	0.0021	—
	TrQA Phi-3	ANS	192	0.710	0.0037	—
	TrQA Phi-3	PH SINK	1024	0.826	0.0010	—

	Setting	Bank	dim	AUC	seed SD	pair acc.
1458						
1459						
1460	TrQA Phi-3	PH 0DTOT	1024	0.810	0.0016	—
1461	TrQA Phi-3	PH 0D	2048	0.807	0.0024	—
1462	TrQA Phi-3	PH ENT	1024	0.817	0.0021	—
1463	TrQA Phi-3	LOOKBACK	1024	0.840	0.0016	—
1464	TrQA Phi-3	HIDDEN	3072	0.867	0.0035	—
1465	TrQA Phi-3	NONTOPO	4198	0.883	0.0030	—
1466	TrQA Phi-3	0D+1D	192	0.705	0.0011	—
1467	TrQA Phi-3	0D+SINK	128	0.713	0.0024	—
1468	TrQA Phi-3	HIDDEN+0D	3136	0.867	0.0037	—
1469	TrQA Phi-3	LEX+0D	—	0.721	0.0031	—
1470	TrQA Phi-3	LEX+DEFL	—	0.712	0.0022	—
1471	TrQA Phi-3	LEX+SINK	—	0.718	0.0044	—
1472	TrQA Phi-3	LOGPROB+0D	70	0.878	0.0013	—
1473	TrQA Phi-3	NONTOPO+0D	4262	0.883	0.0030	—
1474	TrQA Phi-3	NONTOPO+DEFL	4390	0.885	0.0030	—
1475	TrQA Phi-3	PH SINK+PH 0DTOT	2048	0.828	0.0009	—
1476	TrQA Phi-3	SINK+1D	192	0.699	0.0012	—
1477	TrQA Phi-3	SINK+ANS	256	0.736	0.0046	—
1478	TrQA Phi-3	SINK+DEFL	256	0.736	0.0016	—
1479	TrQA TinyLl.	LEN	2	0.631	0.0005	—
1480	TrQA TinyLl.	LEX	—	0.740	0.0003	—
1481	TrQA TinyLl.	LOGPROB	6	0.724	0.0006	—
1482	TrQA TinyLl.	LLMCHECK	22	0.688	0.0010	—
1483	TrQA TinyLl.	ROWSTAT	44	0.752	0.0001	—
1484	TrQA TinyLl.	SINK	44	0.720	0.0020	—
1485	TrQA TinyLl.	0D	44	0.720	0.0020	—
1486	TrQA TinyLl.	LEGACY MSTPROXY	22	0.729	0.0006	—
1487	TrQA TinyLl.	1D	88	0.659	0.0007	—
1488	TrQA TinyLl.	DEFL	132	0.726	0.0024	—
1489	TrQA TinyLl.	ANS	132	0.701	0.0021	—
1490	TrQA TinyLl.	PH SINK	704	0.814	0.0016	—
1491	TrQA TinyLl.	PH 0DTOT	704	0.792	0.0018	—
1492	TrQA TinyLl.	PH 0D	1408	0.775	0.0027	—
1493	TrQA TinyLl.	PH ENT	704	0.807	0.0003	—
1494	TrQA TinyLl.	LOOKBACK	704	0.823	0.0024	—
1495	TrQA TinyLl.	HIDDEN	2048	0.827	0.0028	—
1496	TrQA TinyLl.	NONTOPO	2824	0.842	0.0031	—
1497	TrQA TinyLl.	0D+1D	132	0.718	0.0002	—
1498	TrQA TinyLl.	0D+SINK	88	0.724	0.0015	—
1499	TrQA TinyLl.	HIDDEN+0D	2092	0.828	0.0027	—
1500	TrQA TinyLl.	LEX+0D	—	0.764	0.0010	—
1501	TrQA TinyLl.	LEX+DEFL	—	0.757	0.0032	—
1502	TrQA TinyLl.	LEX+SINK	—	0.762	0.0021	—
1503	TrQA TinyLl.	LOGPROB+0D	50	0.762	0.0021	—
1504	TrQA TinyLl.	NONTOPO+0D	2868	0.842	0.0029	—
1505	TrQA TinyLl.	NONTOPO+DEFL	2956	0.842	0.0032	—
1506	TrQA TinyLl.	PH SINK+PH 0DTOT	1408	0.813	0.0016	—
1507	TrQA TinyLl.	SINK+1D	132	0.719	0.0013	—
1508	TrQA TinyLl.	SINK+ANS	176	0.725	0.0030	—
1509	TrQA TinyLl.	SINK+DEFL	176	0.734	0.0032	—
1510	TrQA Mis7B	LEN	2	0.635	0.0017	—
1511	TrQA Mis7B	LEX	—	0.677	0.0050	—
	TrQA Mis7B	LOGPROB	6	0.719	0.0005	—
	TrQA Mis7B	LLMCHECK	32	0.686	0.0019	—
	TrQA Mis7B	ROWSTAT	64	0.731	0.0016	—
	TrQA Mis7B	SINK	64	0.696	0.0017	—
	TrQA Mis7B	0D	64	0.699	0.0003	—
	TrQA Mis7B	LEGACY MSTPROXY	32	0.693	0.0017	—
	TrQA Mis7B	1D	128	0.611	0.0028	—
	TrQA Mis7B	DEFL	192	0.721	0.0024	—
	TrQA Mis7B	ANS	192	0.641	0.0033	—
	TrQA Mis7B	PH SINK	1024	0.812	0.0024	—
	TrQA Mis7B	PH 0DTOT	1024	0.808	0.0037	—

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Mis7B	PH 0D	2048	0.775	0.0013	—
TrQA Mis7B	PH ENT	1024	0.812	0.0027	—
TrQA Mis7B	LOOKBACK	1024	0.822	0.0035	—
TrQA Mis7B	HIDDEN	4096	0.833	0.0007	—
TrQA Mis7B	NONTOPO	5222	0.841	0.0010	—
TrQA Mis7B	0D+1D	192	0.692	0.0035	—
TrQA Mis7B	0D+SINK	128	0.698	0.0018	—
TrQA Mis7B	HIDDEN+0D	4160	0.834	0.0012	—
TrQA Mis7B	LEX+0D	—	0.723	0.0030	—
TrQA Mis7B	LEX+DEFL	—	0.744	0.0057	—
TrQA Mis7B	LEX+SINK	—	0.721	0.0027	—
TrQA Mis7B	LOGPROB+0D	70	0.752	0.0006	—
TrQA Mis7B	NONTOPO+0D	5286	0.841	0.0015	—
TrQA Mis7B	NONTOPO+DEFL	5414	0.841	0.0011	—
TrQA Mis7B	PH SINK+PH 0DTOT	2048	0.814	0.0022	—
TrQA Mis7B	SINK+1D	192	0.686	0.0042	—
TrQA Mis7B	SINK+ANS	256	0.701	0.0025	—
TrQA Mis7B	SINK+DEFL	256	0.739	0.0019	—
TrQA Qw7B	LEN	2	0.595	0.0004	—
TrQA Qw7B	LEX	—	0.594	0.0029	—
TrQA Qw7B	LOGPROB	6	0.847	0.0003	—
TrQA Qw7B	LLMCHECK	28	0.725	0.0015	—
TrQA Qw7B	ROWSTAT	56	0.734	0.0016	—
TrQA Qw7B	SINK	56	0.714	0.0023	—
TrQA Qw7B	0D	56	0.668	0.0006	—
TrQA Qw7B	LEGACY MSTPROXY	28	0.672	0.0005	—
TrQA Qw7B	1D	112	0.582	0.0022	—
TrQA Qw7B	DEFL	168	0.672	0.0016	—
TrQA Qw7B	ANS	168	0.755	0.0026	—
TrQA Qw7B	PH SINK	784	0.801	0.0009	—
TrQA Qw7B	PH 0DTOT	784	0.822	0.0011	—
TrQA Qw7B	PH 0D	1568	0.798	0.0029	—
TrQA Qw7B	PH ENT	784	0.841	0.0021	—
TrQA Qw7B	LOOKBACK	784	0.879	0.0010	—
TrQA Qw7B	HIDDEN	3584	0.854	0.0007	—
TrQA Qw7B	NONTOPO	4458	0.881	0.0008	—
TrQA Qw7B	0D+1D	168	0.668	0.0021	—
TrQA Qw7B	0D+SINK	112	0.735	0.0008	—
TrQA Qw7B	HIDDEN+0D	3640	0.853	0.0008	—
TrQA Qw7B	LEX+0D	—	0.678	0.0043	—
TrQA Qw7B	LEX+DEFL	—	0.678	0.0022	—
TrQA Qw7B	LEX+SINK	—	0.728	0.0007	—
TrQA Qw7B	LOGPROB+0D	62	0.842	0.0012	—
TrQA Qw7B	NONTOPO+0D	4514	0.881	0.0007	—
TrQA Qw7B	NONTOPO+DEFL	4626	0.881	0.0007	—
TrQA Qw7B	PH SINK+PH 0DTOT	1568	0.832	0.0010	—
TrQA Qw7B	SINK+1D	168	0.718	0.0023	—
TrQA Qw7B	SINK+ANS	224	0.781	0.0035	—
TrQA Qw7B	SINK+DEFL	224	0.734	0.0021	—

Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TQA Qw3B	0D → SINK	0.736	0.737	+0.001	[-0.006, +0.009]	yes	0.90
TQA Qw3B	0D → LEGACY_MSTPROXY	0.736	0.713	-0.023	[-0.034, -0.012]	no	0.42
TQA Qw3B	SINK → 0D+SINK	0.737	0.741	+0.003	[-0.003, +0.010]	yes	0.97
TQA Qw3B	SINK → SINK+DEFL	0.737	0.766	+0.029	[+0.017, +0.041]	no	0.30
TQA Qw3B	0D → 0D+1D	0.736	0.769	+0.033	[+0.022, +0.045]	no	0.38
TQA Qw3B	SINK → SINK+1D	0.737	0.766	+0.028	[+0.017, +0.039]	no	0.46
TQA Qw3B	SINK → SINK+ANS	0.737	0.785	+0.047	[+0.033, +0.062]	no	0.03
TQA Qw3B	PH_0DTOT → PH_SINK	0.842	0.851	+0.009	[+0.002, +0.016]	no	0.93
TQA Qw3B	PH_SINK → PH_SINK+PH_0DTOT	0.851	0.860	+0.009	[+0.005, +0.012]	yes	1.00
TQA Qw3B	PH_0D → PH_ENT	0.853	0.857	+0.004	[-0.006, +0.014]	yes	0.57
TQA Qw3B	0D → LOGPROB	0.736	0.630	-0.106	[-0.129, -0.083]	no	0.00
TQA Qw3B	0D → HIDDEN	0.736	0.913	+0.177	[+0.163, +0.192]	no	0.03
TQA Qw3B	0D → LOOKBACK	0.736	0.897	+0.161	[+0.146, +0.175]	no	0.02
TQA Qw3B	0D → LLMCHECK	0.736	0.693	-0.043	[-0.059, -0.026]	no	0.00

	Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
1566	TQA Qw3B	0D $\rightarrow$ LEX	0.736	0.820	+0.084	[+0.066, +0.102]	no	0.00
1567	TQA Qw3B	0D $\rightarrow$ ROWSTAT	0.736	0.750	+0.014	[+0.002, +0.026]	no	0.30
1568	TQA Qw3B	0D $\rightarrow$ LEN	0.736	0.529	-0.207	[-0.229, -0.185]	no	0.00
1569	TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
1570	TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.926	0.928	+0.001	[-0.000, +0.003]	yes	1.00
1571	TQA Qw3B	LEX $\rightarrow$ LEX+0D	0.820	0.840	+0.020	[+0.011, +0.030]	no	0.63
1572	TQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.820	0.843	+0.022	[+0.013, +0.033]	no	0.59
1573	TQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.820	0.830	+0.010	[-0.003, +0.023]	no	0.19
1574	TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.630	0.776	+0.146	[+0.127, +0.165]	no	0.00
1575	TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.913	0.913	+0.000	[-0.001, +0.001]	yes	1.00
1576	TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
1577	TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
1578	TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.926	0.926	+0.000	[-0.000, +0.001]	yes	1.00
1579	TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.913	0.914	+0.001	[-0.000, +0.002]	yes	1.00
1580	TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.630	0.757	+0.128	[+0.109, +0.147]	no	0.00
1581	TQA Qw3B	SINK $\rightarrow$ SINK(+)0D	0.737	0.739	+0.002	[-0.001, +0.006]	yes	1.00
1582	TQA Qw1.5B	0D $\rightarrow$ SINK	0.726	0.723	-0.003	[-0.012, +0.006]	yes	0.68
1583	TQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.726	0.698	-0.028	[-0.040, -0.016]	no	0.36
1584	TQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.723	0.735	+0.012	[+0.005, +0.020]	no	0.91
1585	TQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.723	0.757	+0.034	[+0.022, +0.045]	no	0.35
1586	TQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.726	0.718	-0.008	[-0.014, -0.002]	yes	0.99
1587	TQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.723	0.715	-0.008	[-0.014, -0.002]	yes	0.99
1588	TQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.723	0.761	+0.038	[+0.024, +0.051]	no	0.11
1589	TQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.819	0.830	+0.011	[+0.003, +0.018]	no	0.90
1590	TQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.830	0.841	+0.011	[+0.007, +0.014]	yes	1.00
1591	TQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.838	0.833	-0.004	[-0.016, +0.007]	no	0.44
1592	TQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.726	0.638	-0.088	[-0.108, -0.067]	no	0.00
1593	TQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.726	0.916	+0.190	[+0.175, +0.204]	no	0.05
1594	TQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.726	0.851	+0.125	[+0.109, +0.140]	no	0.00
1595	TQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.726	0.666	-0.060	[-0.076, -0.044]	no	0.00
1596	TQA Qw1.5B	0D $\rightarrow$ LEX	0.726	0.820	+0.094	[+0.076, +0.112]	no	0.00
1597	TQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.726	0.738	+0.012	[-0.000, +0.025]	no	0.27
1598	TQA Qw1.5B	0D $\rightarrow$ LEN	0.726	0.529	-0.197	[-0.218, -0.176]	no	0.00
1599	TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.917	0.917	+0.000	[-0.000, +0.001]	yes	1.00
1600	TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.917	0.917	+0.000	[-0.001, +0.001]	yes	1.00
1601	TQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.820	0.835	+0.015	[+0.006, +0.024]	no	0.71
1602	TQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.022]	no	0.73
1603	TQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.820	0.825	+0.005	[-0.006, +0.016]	no	0.45
1604	TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.638	0.740	+0.101	[+0.086, +0.116]	no	0.00
1605	TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.916	0.916	+0.001	[-0.000, +0.002]	yes	1.00
1606	TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.916	0.916	+0.000	[-0.001, +0.001]	yes	1.00
1607	TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.916	0.916	-0.000	[-0.001, +0.001]	yes	1.00
1608	TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.916	0.916	-0.000	[-0.001, +0.000]	yes	1.00
1609	TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.915	0.915	+0.000	[-0.001, +0.001]	yes	1.00
1610	TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.638	0.737	+0.100	[+0.084, +0.115]	no	0.00
1611	TQA Qw1.5B	SINK $\rightarrow$ SINK(+)0D	0.722	0.730	+0.007	[+0.002, +0.012]	yes	1.00
1612	TQA Phi-3	0D $\rightarrow$ SINK	0.646	0.651	+0.005	[-0.003, +0.012]	yes	0.90
1613	TQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.646	0.613	-0.033	[-0.046, -0.020]	no	0.20
1614	TQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.651	0.657	+0.006	[-0.000, +0.012]	yes	0.99
1615	TQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.651	0.684	+0.033	[+0.020, +0.048]	no	0.10
1616	TQA Phi-3	0D $\rightarrow$ 0D+1D	0.646	0.643	-0.003	[-0.006, +0.001]	yes	1.00
1617	TQA Phi-3	SINK $\rightarrow$ SINK+1D	0.651	0.651	+0.000	[-0.004, +0.005]	yes	1.00
1618	TQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.651	0.675	+0.024	[+0.007, +0.039]	no	0.00
1619	TQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.845	0.871	+0.026	[+0.020, +0.031]	no	1.00
1620	TQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.871	0.871	-0.000	[-0.001, +0.001]	yes	1.00
1621	TQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.868	0.864	-0.004	[-0.012, +0.004]	yes	0.87
1622	TQA Phi-3	0D $\rightarrow$ LOGPROB	0.646	0.578	-0.068	[-0.092, -0.044]	no	0.00
1623	TQA Phi-3	0D $\rightarrow$ HIDDEN	0.646	0.933	+0.287	[+0.272, +0.302]	no	0.00
1624	TQA Phi-3	0D $\rightarrow$ LOOKBACK	0.646	0.892	+0.246	[+0.230, +0.262]	no	0.00
1625	TQA Phi-3	0D $\rightarrow$ LLMCHECK	0.646	0.653	+0.007	[-0.010, +0.024]	no	0.00
1626	TQA Phi-3	0D $\rightarrow$ LEX	0.646	0.820	+0.174	[+0.154, +0.193]	no	0.00
1627	TQA Phi-3	0D $\rightarrow$ ROWSTAT	0.646	0.661	+0.015	[+0.002, +0.029]	no	0.12
1628	TQA Phi-3	0D $\rightarrow$ LEN	0.646	0.522	-0.124	[-0.145, -0.102]	no	0.00
1629	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.936	0.936	+0.000	[-0.000, +0.001]	yes	1.00
1630	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.936	0.937	+0.001	[-0.000, +0.002]	yes	1.00
1631	TQA Phi-3	LEX $\rightarrow$ LEX+0D	0.820	0.821	+0.001	[-0.007, +0.009]	yes	0.85
1632	TQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.820	0.820	-0.000	[-0.008, +0.007]	yes	0.89
1633	TQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.820	0.802	-0.018	[-0.028, -0.007]	no	0.52
1634	TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.578	0.655	+0.077	[+0.058, +0.096]	no	0.00
1635	TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.933	0.933	+0.000	[-0.001, +0.001]	yes	1.00
1636	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
1637	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
1638	TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.936	0.936	-0.000	[-0.000, +0.000]	yes	1.00
1639	TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.932	0.932	-0.000	[-0.000, -0.000]	yes	1.00
1640	TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.577	0.650	+0.073	[+0.055, +0.091]	no	0.00
1641	TQA Phi-3	SINK $\rightarrow$ SINK(+)0D	0.650	0.651	+0.000	[-0.003, +0.003]	yes	1.00
1642	TQA TinyLl.	0D $\rightarrow$ SINK	0.654	0.643	-0.010	[-0.021, +0.000]	no	0.51
1643	TQA TinyLl.	0D $\rightarrow$ LEGACY_MSTPROXY	0.654	0.627	-0.027	[-0.037, -0.016]	no	0.46
1644	TQA TinyLl.	SINK $\rightarrow$ 0D+SINK	0.643	0.663	+0.020	[+0.011, +0.029]	no	0.73
1645	TQA TinyLl.	SINK $\rightarrow$ SINK+DEFL	0.643	0.688	+0.045	[+0.032, +0.057]	no	0.28
1646	TQA TinyLl.	0D $\rightarrow$ 0D+1D	0.654	0.660	+0.007	[-0.000, +0.013]	yes	0.95
1647	TQA TinyLl.	SINK $\rightarrow$ SINK+1D	0.643	0.651	+0.008	[+0.001, +0.015]	no	0.93
1648	TQA TinyLl.	SINK $\rightarrow$ SINK+ANS	0.643	0.685	+0.041	[+0.025, +0.058]	no	0.00
1649	TQA TinyLl.	PH_0DTOT $\rightarrow$ PH_SINK	0.804	0.808	+0.004	[-0.002, +0.011]	yes	0.97
1650	TQA TinyLl.	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.808	0.810	+0.003	[+0.000, +0.005]	yes	1.00
1651	TQA TinyLl.	PH_0D $\rightarrow$ PH_ENT	0.806	0.817	+0.011	[+0.000, +0.022]	no	0.50
1652	TQA TinyLl.	0D $\rightarrow$ LOGPROB	0.654	0.637	-0.017	[-0.039, +0.004]	no	0.00
1653	TQA TinyLl.	0D $\rightarrow$ HIDDEN	0.654	0.883	+0.229	[+0.214, +0.244]	no	0.00
1654	TQA TinyLl.	0D $\rightarrow$ LOOKBACK	0.654	0.850	+0.197	[+0.182, +0.212]	no	0.00
1655	TQA TinyLl.	0D $\rightarrow$ LLMCHECK	0.654	0.614	-0.039	[-0.055, -0.024]	no	0.00

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TQA TinyLl.	0D $\rightarrow$ LEX	0.654	0.820	+0.166	[+0.149, +0.184]	no	0.00
TQA TinyLl.	0D $\rightarrow$ ROWSTAT	0.654	0.661	+0.007	[-0.004, +0.019]	no	0.37
TQA TinyLl.	0D $\rightarrow$ LEN	0.654	0.523	-0.131	[-0.152, -0.110]	no	0.00
TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+0D	0.885	0.885	+0.000	[-0.000, +0.000]	yes	1.00
TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.885	0.884	-0.001	[-0.002, +0.000]	yes	1.00
TQA TinyLl.	LEX $\rightarrow$ LEX+0D	0.820	0.830	+0.010	[+0.003, +0.016]	no	0.97
TQA TinyLl.	LEX $\rightarrow$ LEX+SINK	0.820	0.831	+0.011	[+0.005, +0.017]	no	0.98
TQA TinyLl.	LEX $\rightarrow$ LEX+DEFL	0.820	0.826	+0.005	[-0.003, +0.014]	yes	0.75
TQA TinyLl.	LOGPROB $\rightarrow$ LOGPROB+0D	0.637	0.683	+0.046	[+0.033, +0.060]	no	0.13
TQA TinyLl.	HIDDEN $\rightarrow$ HIDDEN+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.884	0.884	-0.000	[-0.001, +0.000]	yes	1.00
TQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
TQA TinyLl.	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.883	0.882	-0.000	[-0.000, +0.000]	yes	1.00
TQA TinyLl.	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.636	0.678	+0.043	[+0.030, +0.055]	no	0.25
TQA TinyLl.	SINK $\rightarrow$ SINK(+)0D	0.643	0.657	+0.014	[+0.007, +0.021]	no	0.96
TQA SmolLM	0D $\rightarrow$ SINK	0.583	0.611	+0.028	[+0.009, +0.048]	no	0.00
TQA SmolLM	0D $\rightarrow$ LEGACY_MSTPROXY	0.583	0.585	+0.002	[-0.003, +0.006]	yes	1.00
TQA SmolLM	SINK $\rightarrow$ 0D+SINK	0.611	0.641	+0.030	[+0.021, +0.039]	no	0.66
TQA SmolLM	SINK $\rightarrow$ SINK+DEFL	0.611	0.652	+0.041	[+0.030, +0.053]	no	0.34
TQA SmolLM	0D $\rightarrow$ 0D+1D	0.583	0.588	+0.005	[-0.004, +0.014]	yes	0.68
TQA SmolLM	SINK $\rightarrow$ SINK+1D	0.611	0.629	+0.018	[+0.008, +0.028]	no	0.62
TQA SmolLM	SINK $\rightarrow$ SINK+ANS	0.611	0.646	+0.036	[+0.018, +0.053]	no	0.00
TQA SmolLM	PH_0DTOT $\rightarrow$ PH_SINK	0.819	0.720	-0.099	[-0.113, -0.086]	no	0.15
TQA SmolLM	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.720	0.826	+0.106	[+0.094, +0.118]	no	0.29
TQA SmolLM	PH_0D $\rightarrow$ PH_ENT	0.800	0.834	+0.034	[+0.024, +0.044]	no	0.60
TQA SmolLM	0D $\rightarrow$ LOGPROB	0.583	0.588	+0.005	[-0.017, +0.027]	no	0.00
TQA SmolLM	0D $\rightarrow$ HIDDEN	0.583	0.894	+0.311	[+0.296, +0.326]	no	0.01
TQA SmolLM	0D $\rightarrow$ LOOKBACK	0.583	0.863	+0.280	[+0.264, +0.295]	no	0.00
TQA SmolLM	0D $\rightarrow$ LLMCHECK	0.583	0.604	+0.021	[+0.006, +0.037]	no	0.00
TQA SmolLM	0D $\rightarrow$ LEX	0.583	0.820	+0.237	[+0.218, +0.256]	no	0.00
TQA SmolLM	0D $\rightarrow$ ROWSTAT	0.583	0.673	+0.090	[+0.079, +0.102]	no	0.37
TQA SmolLM	0D $\rightarrow$ LEN	0.583	0.524	-0.059	[-0.074, -0.043]	no	0.00
TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO+0D	0.899	0.899	-0.000	[-0.001, +0.001]	yes	1.00
TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.899	0.899	-0.001	[-0.002, +0.000]	yes	1.00
TQA SmolLM	LEX $\rightarrow$ LEX+0D	0.820	0.827	+0.007	[+0.001, +0.012]	yes	1.00
TQA SmolLM	LEX $\rightarrow$ LEX+SINK	0.820	0.819	-0.001	[-0.007, +0.004]	yes	1.00
TQA SmolLM	LEX $\rightarrow$ LEX+DEFL	0.820	0.822	+0.002	[-0.005, +0.009]	yes	0.91
TQA SmolLM	LOGPROB $\rightarrow$ LOGPROB+0D	0.588	0.613	+0.025	[+0.015, +0.035]	no	0.59
TQA SmolLM	HIDDEN $\rightarrow$ HIDDEN+0D	0.894	0.894	+0.000	[-0.001, +0.001]	yes	1.00
TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
TQA SmolLM	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.899	0.899	-0.000	[-0.000, +0.000]	yes	1.00
TQA SmolLM	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.893	0.893	-0.000	[-0.001, -0.000]	yes	1.00
TQA SmolLM	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.588	0.607	+0.019	[+0.010, +0.029]	no	0.67
TQA SmolLM	SINK $\rightarrow$ SINK(+)0D	0.610	0.633	+0.022	[+0.014, +0.031]	no	0.81
TQA Mis7B	0D $\rightarrow$ SINK	0.684	0.685	+0.000	[-0.001, +0.001]	yes	1.00
TQA Mis7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.684	0.646	-0.038	[-0.051, -0.026]	no	0.24
TQA Mis7B	SINK $\rightarrow$ 0D+SINK	0.685	0.685	+0.000	[-0.001, +0.002]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+DEFL	0.685	0.740	+0.055	[+0.040, +0.070]	no	0.00
TQA Mis7B	0D $\rightarrow$ 0D+1D	0.684	0.685	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+1D	0.685	0.685	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+ANS	0.685	0.727	+0.043	[+0.027, +0.058]	no	0.00
TQA Mis7B	PH_0DTOT $\rightarrow$ PH_SINK	0.887	0.903	+0.016	[+0.011, +0.020]	no	1.00
TQA Mis7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.903	0.903	+0.001	[-0.000, +0.002]	yes	1.00
TQA Mis7B	PH_0D $\rightarrow$ PH_ENT	0.893	0.897	+0.004	[-0.003, +0.011]	yes	0.94
TQA Mis7B	0D $\rightarrow$ LOGPROB	0.684	0.641	-0.044	[-0.067, -0.020]	no	0.00
TQA Mis7B	0D $\rightarrow$ HIDDEN	0.684	0.956	+0.271	[+0.256, +0.286]	no	0.00
TQA Mis7B	0D $\rightarrow$ LOOKBACK	0.684	0.933	+0.248	[+0.233, +0.263]	no	0.00
TQA Mis7B	0D $\rightarrow$ LLMCHECK	0.684	0.652	-0.032	[-0.048, -0.016]	no	0.00
TQA Mis7B	0D $\rightarrow$ LEX	0.684	0.820	+0.136	[+0.116, +0.155]	no	0.00
TQA Mis7B	0D $\rightarrow$ ROWSTAT	0.684	0.694	+0.010	[-0.004, +0.024]	no	0.08
TQA Mis7B	0D $\rightarrow$ LEN	0.684	0.525	-0.159	[-0.179, -0.140]	no	0.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.959	0.958	-0.000	[-0.000, -0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.959	0.959	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	LEX $\rightarrow$ LEX+0D	0.820	0.834	+0.013	[+0.004, +0.023]	no	0.66
TQA Mis7B	LEX $\rightarrow$ LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.023]	no	0.66
TQA Mis7B	LEX $\rightarrow$ LEX+DEFL	0.820	0.823	+0.003	[-0.009, +0.015]	no	0.29
TQA Mis7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.641	0.722	+0.081	[+0.066, +0.097]	no	0.00
TQA Mis7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.956	0.956	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.955	0.955	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.640	0.719	+0.079	[+0.064, +0.094]	no	0.00
TQA Mis7B	SINK $\rightarrow$ SINK(+)0D	0.684	0.684	+0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	0D $\rightarrow$ SINK	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	0D $\rightarrow$ LEGACY_MSTPROXY	0.688	0.652	-0.037	[-0.049, -0.024]	no	0.23
TQA Mis7B (chat)	SINK $\rightarrow$ 0D+SINK	0.688	0.689	+0.000	[-0.001, +0.001]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+DEFL	0.688	0.756	+0.067	[+0.052, +0.082]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ 0D+1D	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+1D	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+ANS	0.688	0.751	+0.063	[+0.048, +0.078]	no	0.01
TQA Mis7B (chat)	PH_0DTOT $\rightarrow$ PH_SINK	0.890	0.907	+0.016	[+0.012, +0.021]	no	1.00
TQA Mis7B (chat)	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.907	0.907	+0.001	[-0.001, +0.002]	yes	1.00
TQA Mis7B (chat)	PH_0D $\rightarrow$ PH_ENT	0.900	0.901	+0.001	[-0.006, +0.008]	yes	0.95
TQA Mis7B (chat)	0D $\rightarrow$ LOGPROB	0.688	0.643	-0.045	[-0.070, -0.021]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ HIDDEN	0.688	0.958	+0.270	[+0.255, +0.284]	no	0.03
TQA Mis7B (chat)	0D $\rightarrow$ LOOKBACK	0.688	0.930	+0.242	[+0.227, +0.256]	no	0.03
TQA Mis7B (chat)	0D $\rightarrow$ LLMCHECK	0.688	0.653	-0.036	[-0.052, -0.019]	no	0.00

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡ _{.015}	power
1674	TQA Mis7B (chat)	0D→LEX	0.688	0.820	+0.132	[+0.112, +0.151]	no	0.00
1675	TQA Mis7B (chat)	0D→ROWSTAT	0.688	0.700	+0.011	[-0.003, +0.025]	no	0.09
1676	TQA Mis7B (chat)	0D→LEN	0.688	0.525	-0.163	[-0.184, -0.143]	no	0.00
1677	TQA Mis7B (chat)	NONTOPO→NONTOPO+0D	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
1678	TQA Mis7B (chat)	NONTOPO→NONTOPO+DEFL	0.960	0.961	+0.000	[-0.000, +0.001]	yes	1.00
1679	TQA Mis7B (chat)	LEX→LEX+0D	0.820	0.834	+0.014	[+0.004, +0.023]	no	0.65
1680	TQA Mis7B (chat)	LEX→LEX+SINK	0.820	0.834	+0.014	[+0.004, +0.023]	no	0.65
1681	TQA Mis7B (chat)	LEX→LEX+DEFL	0.820	0.835	+0.015	[+0.002, +0.027]	no	0.25
1682	TQA Mis7B (chat)	LOGPROB→LOGPROB+0D	0.643	0.729	+0.086	[+0.070, +0.102]	no	0.00
1683	TQA Mis7B (chat)	HIDDEN→HIDDEN+0D	0.958	0.959	+0.000	[-0.000, +0.000]	yes	1.00
1684	TQA Mis7B (chat)	NONTOPO→NONTOPO(+)+0D	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
1685	TQA Mis7B (chat)	NONTOPO→NONTOPO(+)+DEFL	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
1686	TQA Mis7B (chat)	NONTOPO→NONTOPO(+)+SINK	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
1687	TQA Mis7B (chat)	HIDDEN→HIDDEN(+)+0D	0.958	0.958	-0.000	[-0.000, -0.000]	yes	1.00
1688	TQA Mis7B (chat)	LOGPROB→LOGPROB(+)+0D	0.643	0.725	+0.082	[+0.066, +0.098]	no	0.00
1689	TQA Mis7B (chat)	SINK→SINK(+)+0D	0.688	0.688	+0.000	[-0.000, +0.000]	yes	1.00
1690	TQA Qw7B	0D→SINK	0.679	0.749	+0.070	[+0.050, +0.089]	no	0.00
1691	TQA Qw7B	0D→LEGACY_MSTPROXY	0.679	0.678	-0.001	[-0.007, +0.004]	yes	0.99
1692	TQA Qw7B	SINK→0D+SINK	0.749	0.777	+0.027	[+0.021, +0.034]	no	0.95
1693	TQA Qw7B	SINK→SINK+DEFL	0.749	0.788	+0.038	[+0.029, +0.048]	no	0.64
1694	TQA Qw7B	0D→0D+1D	0.679	0.701	+0.021	[+0.011, +0.032]	no	0.52
1695	TQA Qw7B	SINK→SINK+1D	0.749	0.767	+0.017	[+0.010, +0.025]	no	0.89
1696	TQA Qw7B	SINK→SINK+ANS	0.749	0.794	+0.045	[+0.030, +0.060]	no	0.02
1697	TQA Qw7B	PH_0DTOT→PH_SINK	0.872	0.821	-0.051	[-0.062, -0.040]	no	0.45
1698	TQA Qw7B	PH_SINK→PH_SINK+PH_0DTOT	0.821	0.881	+0.060	[+0.051, +0.069]	no	0.71
1699	TQA Qw7B	PH_0D→PH_ENT	0.850	0.887	+0.037	[+0.027, +0.048]	no	0.52
1700	TQA Qw7B	0D→LOGPROB	0.679	0.674	-0.006	[-0.028, +0.017]	no	0.00
1701	TQA Qw7B	0D→HIDDEN	0.679	0.933	+0.254	[+0.240, +0.268]	no	0.09
1702	TQA Qw7B	0D→LOOKBACK	0.679	0.922	+0.242	[+0.229, +0.256]	no	0.17
1703	TQA Qw7B	0D→LLMCHECK	0.679	0.667	-0.012	[-0.027, +0.004]	no	0.00
1704	TQA Qw7B	0D→LEX	0.679	0.820	+0.141	[+0.122, +0.159]	no	0.00
1705	TQA Qw7B	0D→ROWSTAT	0.679	0.736	+0.057	[+0.048, +0.066]	no	0.74
1706	TQA Qw7B	0D→LEN	0.679	0.529	-0.151	[-0.171, -0.130]	no	0.00
1707	TQA Qw7B	NONTOPO→NONTOPO+0D	0.950	0.949	-0.000	[-0.000, +0.000]	yes	1.00
1708	TQA Qw7B	NONTOPO→NONTOPO+DEFL	0.950	0.949	-0.000	[-0.001, +0.000]	yes	1.00
1709	TQA Qw7B	LEX→LEX+0D	0.820	0.834	+0.014	[+0.007, +0.021]	no	0.94
1710	TQA Qw7B	LEX→LEX+SINK	0.820	0.848	+0.028	[+0.016, +0.040]	no	0.29
1711	TQA Qw7B	LEX→LEX+DEFL	0.820	0.828	+0.008	[-0.001, +0.016]	no	0.78
1712	TQA Qw7B	LOGPROB→LOGPROB+0D	0.674	0.727	+0.053	[+0.042, +0.065]	no	0.40
1713	TQA Qw7B	HIDDEN→HIDDEN+0D	0.933	0.933	-0.000	[-0.001, +0.000]	yes	1.00
1714	TQA Qw7B	NONTOPO→NONTOPO(+)+0D	0.949	0.949	-0.000	[-0.000, +0.000]	yes	1.00
1715	TQA Qw7B	NONTOPO→NONTOPO(+)+DEFL	0.949	0.949	+0.000	[-0.000, +0.000]	yes	1.00
1716	TQA Qw7B	NONTOPO→NONTOPO(+)+SINK	0.949	0.949	+0.000	[-0.001, +0.001]	yes	1.00
1717	TQA Qw7B	HIDDEN→HIDDEN(+)+0D	0.933	0.933	+0.000	[-0.000, +0.001]	yes	1.00
1718	TQA Qw7B	LOGPROB→LOGPROB(+)+0D	0.673	0.725	+0.052	[+0.039, +0.064]	no	0.23
1719	TQA Qw7B	SINK→SINK(+)+0D	0.749	0.775	+0.026	[+0.018, +0.034]	no	0.88
1720	HE Qw3B	0D→SINK	0.811	0.829	+0.018	[+0.008, +0.028]	no	0.64
1721	HE Qw3B	0D→LEGACY_MSTPROXY	0.811	0.731	-0.080	[-0.091, -0.069]	no	0.46
1722	HE Qw3B	SINK→0D+SINK	0.829	0.857	+0.028	[+0.022, +0.035]	no	0.97
1723	HE Qw3B	SINK→SINK+DEFL	0.829	0.887	+0.058	[+0.049, +0.067]	no	0.73
1724	HE Qw3B	0D→0D+1D	0.811	0.826	+0.015	[+0.009, +0.023]	no	0.94
1725	HE Qw3B	SINK→SINK+1D	0.829	0.843	+0.014	[+0.007, +0.021]	no	0.97
1726	HE Qw3B	SINK→SINK+ANS	0.829	0.978	+0.150	[+0.139, +0.161]	no	0.44
1727	HE Qw3B	PH_0DTOT→PH_SINK	0.958	0.958	+0.000	[-0.005, +0.006]	yes	1.00
1728	HE Qw3B	PH_SINK→PH_SINK+PH_0DTOT	0.958	0.962	+0.004	[+0.003, +0.005]	yes	1.00
1729	HE Qw3B	PH_0D→PH_ENT	0.971	0.964	-0.006	[-0.012, -0.000]	yes	0.99
1730	HE Qw3B	0D→LOGPROB	0.811	0.993	+0.182	[+0.170, +0.194]	no	0.33
1731	HE Qw3B	0D→HIDDEN	0.811	1.000	+0.189	[+0.177, +0.201]	no	0.32
1732	HE Qw3B	0D→LOOKBACK	0.811	0.998	+0.187	[+0.175, +0.199]	no	0.32
1733	HE Qw3B	0D→LLMCHECK	0.811	0.745	-0.066	[-0.078, -0.054]	no	0.27
1734	HE Qw3B	0D→LEX	0.811	0.981	+0.170	[+0.159, +0.182]	no	0.37
1735	HE Qw3B	0D→ROWSTAT	0.811	0.829	+0.018	[+0.007, +0.030]	no	0.35
1736	HE Qw3B	0D→LEN	0.811	0.962	+0.151	[+0.140, +0.163]	no	0.36
1737	HE Qw3B	NONTOPO→NONTOPO+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1738	HE Qw3B	NONTOPO→NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1739	HE Qw3B	LEX→LEX+0D	0.981	0.979	-0.002	[-0.005, +0.000]	yes	1.00
1740	HE Qw3B	LEX→LEX+SINK	0.981	0.980	-0.001	[-0.003, +0.001]	yes	1.00
1741	HE Qw3B	LEX→LEX+DEFL	0.981	0.971	-0.010	[-0.014, -0.007]	yes	1.00
1742	HE Qw3B	LOGPROB→LOGPROB+0D	0.993	0.993	+0.000	[-0.001, +0.001]	yes	1.00
1743	HE Qw3B	HIDDEN→HIDDEN+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1744	HE Qw3B	NONTOPO→NONTOPO(+)+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1745	HE Qw3B	NONTOPO→NONTOPO(+)+DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1746	HE Qw3B	NONTOPO→NONTOPO(+)+SINK	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1747	HE Qw3B	HIDDEN→HIDDEN(+)+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1748	HE Qw3B	LOGPROB→LOGPROB(+)+0D	0.993	0.993	+0.001	[+0.000, +0.001]	yes	1.00
1749	HE Qw3B	SINK→SINK(+)+0D	0.828	0.839	+0.011	[+0.007, +0.015]	yes	1.00
1750	HE Qw1.5B	0D→SINK	0.789	0.820	+0.031	[+0.017, +0.045]	no	0.08
1751	HE Qw1.5B	0D→LEGACY_MSTPROXY	0.789	0.719	-0.070	[-0.086, -0.053]	no	0.00
1752	HE Qw1.5B	SINK→0D+SINK	0.820	0.832	+0.012	[+0.005, +0.019]	no	0.91
1753	HE Qw1.5B	SINK→SINK+DEFL	0.820	0.872	+0.052	[+0.038, +0.065]	no	0.17
1754	HE Qw1.5B	0D→0D+1D	0.789	0.815	+0.026	[+0.013, +0.040]	no	0.17
1755	HE Qw1.5B	SINK→SINK+1D	0.820	0.826	+0.006	[-0.005, +0.017]	no	0.45
1756	HE Qw1.5B	SINK→SINK+ANS	0.820	0.975	+0.155	[+0.138, +0.172]	no	0.00
1757	HE Qw1.5B	PH_0DTOT→PH_SINK	0.921	0.922	+0.002	[-0.009, +0.013]	yes	0.46
1758	HE Qw1.5B	PH_SINK→PH_SINK+PH_0DTOT	0.922	0.931	+0.008	[+0.006, +0.011]	yes	1.00
1759	HE Qw1.5B	PH_0D→PH_ENT	0.949	0.933	-0.016	[-0.028, -0.004]	no	0.33
1760	HE Qw1.5B	0D→LOGPROB	0.789	0.995	+0.205	[+0.188, +0.222]	no	0.00
1761	HE Qw1.5B	0D→HIDDEN	0.789	1.000	+0.210	[+0.193, +0.227]	no	0.00
1762	HE Qw1.5B	0D→LOOKBACK	0.789	0.999	+0.210	[+0.193, +0.226]	no	0.00
1763	HE Qw1.5B	0D→LLMCHECK	0.789	0.791	+0.002	[-0.017, +0.021]	no	0.00

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡ _{.015}	power
1728	HE Qw1.5B	0D→LEX	0.789	0.979	+0.190	[+0.172, +0.207]	no	0.00
1729	HE Qw1.5B	0D→ROWSTAT	0.789	0.788	-0.001	[-0.018, +0.016]	no	0.00
1730	HE Qw1.5B	0D→LEN	0.789	0.967	+0.178	[+0.160, +0.196]	no	0.00
1731	HE Qw1.5B	NONTOPO→NONTOPO+0D	1.000	1.000	+0.000	[+0.000, +0.000]	yes	1.00
1732	HE Qw1.5B	NONTOPO→NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1733	HE Qw1.5B	LEX→LEX+0D	0.979	0.971	-0.008	[-0.014, -0.002]	yes	0.99
1733	HE Qw1.5B	LEX→LEX+SINK	0.979	0.968	-0.011	[-0.017, -0.005]	no	0.98
1734	HE Qw1.5B	LEX→LEX+DEFL	0.979	0.956	-0.022	[-0.030, -0.016]	no	0.94
1734	HE Qw1.5B	LOGPROB→LOGPROB+0D	0.995	0.994	-0.001	[-0.003, +0.001]	yes	1.00
1735	HE Qw1.5B	HIDDEN→HIDDEN+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1735	HE Qw1.5B	NONTOPO→NONTOPO(+)+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1736	HE Qw1.5B	NONTOPO→NONTOPO(+)+DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1736	HE Qw1.5B	NONTOPO→NONTOPO(+)+SINK	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
1737	HE Qw1.5B	HIDDEN→HIDDEN(+)+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
1738	HE Qw1.5B	LOGPROB→LOGPROB(+)+0D	0.994	0.994	-0.000	[-0.001, +0.001]	yes	1.00
1738	HE Qw1.5B	SINK→SINK(+)+0D	0.819	0.823	+0.004	[-0.000, +0.009]	yes	1.00
1739	TrQA Qw3B	0D→SINK	0.732	0.717	-0.015	[-0.026, -0.004]	no	0.48
1739	TrQA Qw3B	0D→LEGACY_MSTPROXY	0.732	0.691	-0.041	[-0.057, -0.025]	no	0.00
1740	TrQA Qw3B	SINK→0D+SINK	0.717	0.738	+0.021	[+0.012, +0.030]	no	0.76
1740	TrQA Qw3B	SINK→SINK+DEFL	0.717	0.719	+0.002	[-0.012, +0.015]	no	0.13
1741	TrQA Qw3B	0D→0D+1D	0.732	0.729	-0.003	[-0.012, +0.006]	yes	0.74
1742	TrQA Qw3B	SINK→SINK+1D	0.717	0.718	+0.001	[-0.009, +0.010]	yes	0.65
1742	TrQA Qw3B	SINK→SINK+ANS	0.717	0.752	+0.035	[+0.019, +0.051]	no	0.00
1743	TrQA Qw3B	PH_0DTOT→PH_SINK	0.807	0.820	+0.012	[+0.004, +0.021]	no	0.82
1743	TrQA Qw3B	PH_SINK→PH_SINK+PH_0DTOT	0.820	0.824	+0.004	[+0.002, +0.006]	yes	1.00
1744	TrQA Qw3B	PH_0D→PH_ENT	0.807	0.810	+0.003	[-0.011, +0.017]	no	0.08
1745	TrQA Qw3B	0D→LOGPROB	0.732	0.839	+0.107	[+0.088, +0.125]	no	0.00
1745	TrQA Qw3B	0D→HIDDEN	0.732	0.841	+0.109	[+0.090, +0.129]	no	0.00
1746	TrQA Qw3B	0D→LOOKBACK	0.732	0.861	+0.129	[+0.111, +0.147]	no	0.00
1746	TrQA Qw3B	0D→LLMCHECK	0.732	0.688	-0.044	[-0.065, -0.023]	no	0.00
1747	TrQA Qw3B	0D→LEX	0.732	0.596	-0.137	[-0.163, -0.110]	no	0.00
1747	TrQA Qw3B	0D→ROWSTAT	0.732	0.724	-0.009	[-0.025, +0.008]	no	0.00
1748	TrQA Qw3B	0D→LEN	0.732	0.569	-0.163	[-0.189, -0.137]	no	0.00
1749	TrQA Qw3B	NONTOPO→NONTOPO+0D	0.870	0.870	+0.001	[-0.000, +0.002]	yes	1.00
1749	TrQA Qw3B	NONTOPO→NONTOPO+DEFL	0.870	0.870	+0.000	[-0.001, +0.002]	yes	1.00
1750	TrQA Qw3B	LEX→LEX+0D	0.596	0.733	+0.137	[+0.117, +0.158]	no	0.00
1750	TrQA Qw3B	LEX→LEX+SINK	0.596	0.724	+0.128	[+0.108, +0.149]	no	0.00
1751	TrQA Qw3B	LEX→LEX+DEFL	0.596	0.690	+0.095	[+0.073, +0.116]	no	0.00
1752	TrQA Qw3B	LOGPROB→LOGPROB+0D	0.839	0.843	+0.004	[-0.002, +0.011]	yes	0.95
1752	TrQA Qw3B	HIDDEN→HIDDEN+0D	0.841	0.843	+0.002	[+0.000, +0.004]	yes	1.00
1753	TrQA Qw3B	NONTOPO→NONTOPO(+)+0D	0.869	0.871	+0.002	[-0.000, +0.004]	yes	1.00
1753	TrQA Qw3B	NONTOPO→NONTOPO(+)+DEFL	0.869	0.870	+0.001	[-0.001, +0.002]	yes	1.00
1754	TrQA Qw3B	NONTOPO→NONTOPO(+)+SINK	0.869	0.871	+0.001	[-0.001, +0.004]	yes	1.00
1754	TrQA Qw3B	HIDDEN→HIDDEN(+)+0D	0.841	0.849	+0.008	[+0.003, +0.012]	yes	1.00
1755	TrQA Qw3B	LOGPROB→LOGPROB(+)+0D	0.838	0.844	+0.006	[+0.002, +0.010]	yes	1.00
1756	TrQA Qw3B	SINK→SINK(+)+0D	0.716	0.733	+0.017	[+0.010, +0.025]	no	0.89
1756	TrQA Qw1.5B	0D→SINK	0.716	0.707	-0.009	[-0.020, +0.002]	no	0.43
1757	TrQA Qw1.5B	0D→LEGACY_MSTPROXY	0.716	0.685	-0.031	[-0.046, -0.015]	no	0.00
1758	TrQA Qw1.5B	SINK→0D+SINK	0.707	0.717	+0.010	[+0.001, +0.020]	no	0.68
1758	TrQA Qw1.5B	SINK→SINK+DEFL	0.707	0.719	+0.012	[-0.003, +0.027]	no	0.00
1759	TrQA Qw1.5B	0D→0D+1D	0.716	0.714	-0.002	[-0.011, +0.008]	yes	0.65
1759	TrQA Qw1.5B	SINK→SINK+1D	0.707	0.709	+0.002	[-0.008, +0.013]	yes	0.49
1760	TrQA Qw1.5B	SINK→SINK+ANS	0.707	0.726	+0.019	[+0.001, +0.036]	no	0.00
1760	TrQA Qw1.5B	PH_0DTOT→PH_SINK	0.759	0.767	+0.008	[-0.003, +0.018]	no	0.53
1761	TrQA Qw1.5B	PH_SINK→PH_SINK+PH_0DTOT	0.767	0.772	+0.004	[+0.001, +0.008]	yes	1.00
1761	TrQA Qw1.5B	PH_0D→PH_ENT	0.768	0.768	-0.000	[-0.017, +0.017]	no	0.00
1762	TrQA Qw1.5B	0D→LOGPROB	0.716	0.823	+0.107	[+0.085, +0.129]	no	0.00
1763	TrQA Qw1.5B	0D→HIDDEN	0.716	0.855	+0.140	[+0.119, +0.160]	no	0.00
1763	TrQA Qw1.5B	0D→LOOKBACK	0.716	0.824	+0.108	[+0.088, +0.129]	no	0.00
1764	TrQA Qw1.5B	0D→LLMCHECK	0.716	0.676	-0.039	[-0.062, -0.016]	no	0.00
1764	TrQA Qw1.5B	0D→LEX	0.716	0.609	-0.106	[-0.136, -0.078]	no	0.00
1765	TrQA Qw1.5B	0D→ROWSTAT	0.716	0.720	+0.005	[-0.011, +0.020]	no	0.00
1765	TrQA Qw1.5B	0D→LEN	0.716	0.588	-0.128	[-0.154, -0.102]	no	0.00
1766	TrQA Qw1.5B	NONTOPO→NONTOPO+0D	0.873	0.873	+0.000	[-0.001, +0.001]	yes	1.00
1767	TrQA Qw1.5B	NONTOPO→NONTOPO+DEFL	0.873	0.872	-0.001	[-0.002, +0.001]	yes	1.00
1768	TrQA Qw1.5B	LEX→LEX+0D	0.609	0.721	+0.112	[+0.091, +0.134]	no	0.00
1768	TrQA Qw1.5B	LEX→LEX+SINK	0.609	0.715	+0.106	[+0.085, +0.127]	no	0.00
1769	TrQA Qw1.5B	LEX→LEX+DEFL	0.609	0.672	+0.063	[+0.040, +0.085]	no	0.00
1769	TrQA Qw1.5B	LOGPROB→LOGPROB+0D	0.823	0.828	+0.006	[-0.001, +0.013]	yes	0.92
1770	TrQA Qw1.5B	HIDDEN→HIDDEN+0D	0.855	0.855	+0.000	[-0.001, +0.002]	yes	1.00
1770	TrQA Qw1.5B	NONTOPO→NONTOPO(+)+0D	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
1771	TrQA Qw1.5B	NONTOPO→NONTOPO(+)+DEFL	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
1771	TrQA Qw1.5B	NONTOPO→NONTOPO(+)+SINK	0.872	0.873	+0.001	[-0.001, +0.002]	yes	1.00
1772	TrQA Qw1.5B	HIDDEN→HIDDEN(+)+0D	0.855	0.857	+0.002	[-0.000, +0.004]	yes	1.00
1772	TrQA Qw1.5B	LOGPROB→LOGPROB(+)+0D	0.822	0.828	+0.006	[+0.001, +0.010]	yes	1.00
1773	TrQA Qw1.5B	SINK→SINK(+)+0D	0.706	0.717	+0.011	[+0.003, +0.018]	no	0.93
1774	TrQA Phi-3	0D→SINK	0.710	0.707	-0.004	[-0.010, +0.002]	yes	0.98
1774	TrQA Phi-3	0D→LEGACY_MSTPROXY	0.710	0.669	-0.041	[-0.057, -0.025]	no	0.00
1775	TrQA Phi-3	SINK→0D+SINK	0.707	0.713	+0.006	[+0.000, +0.012]	yes	0.99
1775	TrQA Phi-3	SINK→SINK+DEFL	0.707	0.736	+0.029	[+0.013, +0.045]	no	0.00
1776	TrQA Phi-3	0D→0D+1D	0.710	0.705	-0.005	[-0.011, +0.000]	yes	0.99
1777	TrQA Phi-3	SINK→SINK+1D	0.707	0.699	-0.008	[-0.013, -0.002]	yes	1.00
1777	TrQA Phi-3	SINK→SINK+ANS	0.707	0.736	+0.029	[+0.012, +0.046]	no	0.00
1778	TrQA Phi-3	PH_0DTOT→PH_SINK	0.810	0.826	+0.016	[+0.009, +0.022]	no	0.96
1778	TrQA Phi-3	PH_SINK→PH_SINK+PH_0DTOT	0.826	0.828	+0.002	[-0.001, +0.004]	yes	1.00
1779	TrQA Phi-3	PH_0D→PH_ENT	0.807	0.817	+0.010	[-0.002, +0.022]	no	0.31
1779	TrQA Phi-3	0D→LOGPROB	0.710	0.877	+0.167	[+0.148, +0.186]	no	0.00
1780	TrQA Phi-3	0D→HIDDEN	0.710	0.867	+0.157	[+0.137, +0.176]	no	0.00
1780	TrQA Phi-3	0D→LOOKBACK	0.710	0.840	+0.129	[+0.109, +0.150]	no	0.00
1781	TrQA Phi-3	0D→LLMCHECK	0.710	0.676	-0.034	[-0.055, -0.014]	no	0.00

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡ _{.015}	power
1782	TrQA Phi-3	0D→LEX	0.710	0.635	-0.076	[-0.102, -0.050]	no	0.00
1783	TrQA Phi-3	0D→ROWSTAT	0.710	0.728	+0.018	[+0.002, +0.034]	no	0.00
1784	TrQA Phi-3	0D→LEN	0.710	0.581	-0.130	[-0.156, -0.104]	no	0.00
1785	TrQA Phi-3	NONTOPO→NONTOPO+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
1786	TrQA Phi-3	NONTOPO→NONTOPO+DEFL	0.883	0.885	+0.002	[+0.000, +0.004]	yes	1.00
1787	TrQA Phi-3	LEX→LEX+0D	0.635	0.721	+0.086	[+0.069, +0.104]	no	0.00
1788	TrQA Phi-3	LEX→LEX+SINK	0.635	0.718	+0.083	[+0.066, +0.101]	no	0.00
1789	TrQA Phi-3	LEX→LEX+DEFL	0.635	0.712	+0.077	[+0.057, +0.098]	no	0.00
1790	TrQA Phi-3	LOGPROB→LOGPROB+0D	0.877	0.878	+0.000	[-0.002, +0.002]	yes	1.00
1791	TrQA Phi-3	HIDDEN→HIDDEN+0D	0.867	0.867	-0.000	[-0.001, +0.000]	yes	1.00
1792	TrQA Phi-3	NONTOPO→NONTOPO(+)+0D	0.883	0.884	+0.000	[-0.001, +0.002]	yes	1.00
1793	TrQA Phi-3	NONTOPO→NONTOPO(+)+DEFL	0.883	0.884	+0.001	[-0.001, +0.003]	yes	1.00
1794	TrQA Phi-3	NONTOPO→NONTOPO(+)+SINK	0.883	0.883	+0.000	[-0.001, +0.001]	yes	1.00
1795	TrQA Phi-3	HIDDEN→HIDDEN(+)+0D	0.867	0.869	+0.002	[-0.000, +0.004]	yes	1.00
1796	TrQA Phi-3	LOGPROB→LOGPROB(+)+0D	0.877	0.878	+0.001	[-0.001, +0.003]	yes	1.00
1797	TrQA Phi-3	SINK→SINK(+)+0D	0.706	0.710	+0.004	[-0.001, +0.009]	yes	1.00
1798	TrQA TinyLL	0D→SINK	0.720	0.720	+0.001	[-0.010, +0.012]	yes	0.40
1799	TrQA TinyLL	0D→LEGACY_MSTPROXY	0.720	0.729	+0.010	[-0.000, +0.020]	no	0.55
1800	TrQA TinyLL	SINK→0D+SINK	0.720	0.724	+0.004	[-0.003, +0.011]	yes	0.93
1801	TrQA TinyLL	SINK→SINK+DEFL	0.720	0.734	+0.014	[-0.000, +0.028]	no	0.10
1802	TrQA TinyLL	0D→0D+1D	0.720	0.718	-0.002	[-0.011, +0.008]	yes	0.60
1803	TrQA TinyLL	SINK→SINK+1D	0.720	0.719	-0.001	[-0.010, +0.008]	yes	0.73
1804	TrQA TinyLL	SINK→SINK+ANS	0.720	0.725	+0.005	[-0.011, +0.021]	no	0.00
1805	TrQA TinyLL	PH_0DTOT→PH_SINK	0.792	0.814	+0.022	[+0.015, +0.029]	no	0.94
1806	TrQA TinyLL	PH_SINK→PH_SINK+PH_0DTOT	0.814	0.813	-0.001	[-0.002, +0.000]	yes	1.00
1807	TrQA TinyLL	PH_0D→PH_ENT	0.775	0.807	+0.032	[+0.017, +0.047]	no	0.01
1808	TrQA TinyLL	0D→LOGPROB	0.720	0.724	+0.004	[-0.019, +0.027]	no	0.00
1809	TrQA TinyLL	0D→HIDDEN	0.720	0.827	+0.108	[+0.088, +0.128]	no	0.00
1810	TrQA TinyLL	0D→LOOKBACK	0.720	0.823	+0.104	[+0.087, +0.121]	no	0.00
1811	TrQA TinyLL	0D→LLMCHECK	0.720	0.688	-0.032	[-0.050, -0.013]	no	0.00
1812	TrQA TinyLL	0D→LEX	0.720	0.740	+0.021	[-0.003, +0.044]	no	0.00
1813	TrQA TinyLL	0D→ROWSTAT	0.720	0.752	+0.032	[+0.020, +0.045]	no	0.26
1814	TrQA TinyLL	0D→LEN	0.720	0.631	-0.089	[-0.109, -0.068]	no	0.00
1815	TrQA TinyLL	NONTOPO→NONTOPO+0D	0.842	0.842	-0.000	[-0.001, +0.001]	yes	1.00
1816	TrQA TinyLL	NONTOPO→NONTOPO+DEFL	0.842	0.842	+0.000	[-0.003, +0.003]	yes	1.00
1817	TrQA TinyLL	LEX→LEX+0D	0.740	0.764	+0.024	[+0.012, +0.036]	no	0.33
1818	TrQA TinyLL	LEX→LEX+SINK	0.740	0.762	+0.021	[+0.010, +0.032]	no	0.45
1819	TrQA TinyLL	LEX→LEX+DEFL	0.740	0.757	+0.016	[+0.002, +0.031]	no	0.04
1820	TrQA TinyLL	LOGPROB→LOGPROB+0D	0.724	0.762	+0.038	[+0.025, +0.052]	no	0.17
1821	TrQA TinyLL	HIDDEN→HIDDEN+0D	0.827	0.828	+0.001	[-0.001, +0.002]	yes	1.00
1822	TrQA TinyLL	NONTOPO→NONTOPO(+)+0D	0.842	0.842	+0.000	[-0.001, +0.002]	yes	1.00
1823	TrQA TinyLL	NONTOPO→NONTOPO(+)+DEFL	0.842	0.843	+0.001	[-0.001, +0.004]	yes	1.00
1824	TrQA TinyLL	NONTOPO→NONTOPO(+)+SINK	0.842	0.842	+0.001	[-0.001, +0.002]	yes	1.00
1825	TrQA TinyLL	HIDDEN→HIDDEN(+)+0D	0.827	0.828	+0.001	[-0.001, +0.004]	yes	1.00
1826	TrQA TinyLL	LOGPROB→LOGPROB(+)+0D	0.723	0.755	+0.031	[+0.020, +0.043]	no	0.40
1827	TrQA TinyLL	SINK→SINK(+)+0D	0.720	0.725	+0.005	[+0.001, +0.009]	yes	1.00
1828	TrQA Mis7B	0D→SINK	0.699	0.696	-0.003	[-0.013, +0.007]	yes	0.62
1829	TrQA Mis7B	0D→LEGACY_MSTPROXY	0.699	0.693	-0.006	[-0.019, +0.007]	no	0.21
1830	TrQA Mis7B	SINK→0D+SINK	0.696	0.698	+0.002	[-0.007, +0.010]	yes	0.80
1831	TrQA Mis7B	SINK→SINK+DEFL	0.696	0.739	+0.043	[+0.026, +0.060]	no	0.00
1832	TrQA Mis7B	0D→0D+1D	0.699	0.692	-0.007	[-0.017, +0.003]	no	0.59
1833	TrQA Mis7B	SINK→SINK+1D	0.696	0.686	-0.010	[-0.020, +0.001]	no	0.52
1834	TrQA Mis7B	SINK→SINK+ANS	0.696	0.701	+0.005	[-0.010, +0.020]	no	0.00
1835	TrQA Mis7B	PH_0DTOT→PH_SINK	0.808	0.812	+0.004	[-0.004, +0.011]	yes	0.88
1836	TrQA Mis7B	PH_SINK→PH_SINK+PH_0DTOT	0.812	0.814	+0.002	[+0.000, +0.004]	yes	1.00
1837	TrQA Mis7B	PH_0D→PH_ENT	0.775	0.812	+0.037	[+0.023, +0.051]	no	0.09
1838	TrQA Mis7B	0D→LOGPROB	0.699	0.719	+0.020	[-0.005, +0.044]	no	0.00
1839	TrQA Mis7B	0D→HIDDEN	0.699	0.833	+0.134	[+0.113, +0.155]	no	0.00
1840	TrQA Mis7B	0D→LOOKBACK	0.699	0.822	+0.123	[+0.103, +0.143]	no	0.00
1841	TrQA Mis7B	0D→LLMCHECK	0.699	0.686	-0.013	[-0.034, +0.008]	no	0.00
1842	TrQA Mis7B	0D→LEX	0.699	0.677	-0.023	[-0.047, +0.002]	no	0.00
1843	TrQA Mis7B	0D→ROWSTAT	0.699	0.731	+0.032	[+0.017, +0.047]	no	0.03
1844	TrQA Mis7B	0D→LEN	0.699	0.635	-0.064	[-0.086, -0.043]	no	0.00
1845	TrQA Mis7B	NONTOPO→NONTOPO+0D	0.841	0.841	+0.000	[-0.001, +0.001]	yes	1.00
1846	TrQA Mis7B	NONTOPO→NONTOPO+DEFL	0.841	0.841	-0.000	[-0.001, +0.001]	yes	1.00
1847	TrQA Mis7B	LEX→LEX+0D	0.677	0.723	+0.046	[+0.030, +0.062]	no	0.00
1848	TrQA Mis7B	LEX→LEX+SINK	0.677	0.721	+0.045	[+0.029, +0.060]	no	0.00
1849	TrQA Mis7B	LEX→LEX+DEFL	0.677	0.744	+0.068	[+0.049, +0.087]	no	0.00
1850	TrQA Mis7B	LOGPROB→LOGPROB+0D	0.719	0.752	+0.033	[+0.017, +0.050]	no	0.00
1851	TrQA Mis7B	HIDDEN→HIDDEN+0D	0.833	0.834	+0.001	[-0.001, +0.002]	yes	1.00
1852	TrQA Mis7B	NONTOPO→NONTOPO(+)+0D	0.841	0.843	+0.003	[+0.000, +0.005]	yes	1.00
1853	TrQA Mis7B	NONTOPO→NONTOPO(+)+DEFL	0.841	0.846	+0.006	[+0.002, +0.009]	yes	1.00
1854	TrQA Mis7B	NONTOPO→NONTOPO(+)+SINK	0.841	0.843	+0.003	[-0.000, +0.005]	yes	1.00
1855	TrQA Mis7B	HIDDEN→HIDDEN(+)+0D	0.833	0.837	+0.004	[+0.001, +0.006]	yes	1.00
1856	TrQA Mis7B	LOGPROB→LOGPROB(+)+0D	0.718	0.749	+0.031	[+0.016, +0.046]	no	0.00
1857	TrQA Mis7B	SINK→SINK(+)+0D	0.696	0.700	+0.005	[-0.001, +0.011]	yes	0.98
1858	TrQA Qw7B	0D→SINK	0.668	0.714	+0.046	[+0.022, +0.070]	no	0.00
1859	TrQA Qw7B	0D→LEGACY_MSTPROXY	0.668	0.672	+0.004	[-0.002, +0.010]	yes	0.98
1860	TrQA Qw7B	SINK→0D+SINK	0.714	0.735	+0.022	[+0.011, +0.033]	no	0.46
1861	TrQA Qw7B	SINK→SINK+DEFL	0.714	0.734	+0.020	[+0.008, +0.033]	no	0.27
1862	TrQA Qw7B	0D→0D+1D	0.668	0.668	+0.000	[-0.010, +0.010]	yes	0.59
1863	TrQA Qw7B	SINK→SINK+1D	0.714	0.718	+0.004	[-0.004, +0.013]	yes	0.73
1864	TrQA Qw7B	SINK→SINK+ANS	0.714	0.781	+0.067	[+0.051, +0.083]	no	0.00
1865	TrQA Qw7B	PH_0DTOT→PH_SINK	0.822	0.801	-0.020	[-0.035, -0.006]	no	0.07
1866	TrQA Qw7B	PH_SINK→PH_SINK+PH_0DTOT	0.801	0.832	+0.031	[+0.024, +0.039]	no	0.91
1867	TrQA Qw7B	PH_0D→PH_ENT	0.798	0.841	+0.044	[+0.032, +0.056]	no	0.33
1868	TrQA Qw7B	0D→LOGPROB	0.668	0.847	+0.180	[+0.158, +0.201]	no	0.00
1869	TrQA Qw7B	0D→HIDDEN	0.668	0.854	+0.186	[+0.165, +0.207]	no	0.00
1870	TrQA Qw7B	0D→LOOKBACK	0.668	0.879	+0.211	[+0.191, +0.231]	no	0.00
1871	TrQA Qw7B	0D→LLMCHECK	0.668	0.725	+0.058	[+0.038, +0.078]	no	0.00

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\Xi_{.015}$	power
TrQA Qw7B	0D $\rightarrow$ LEX	0.668	0.594	-0.074	[-0.101, -0.046]	no	0.00
TrQA Qw7B	0D $\rightarrow$ ROWSTAT	0.668	0.734	+0.067	[+0.053, +0.081]	no	0.10
TrQA Qw7B	0D $\rightarrow$ LEN	0.668	0.595	-0.073	[-0.097, -0.049]	no	0.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.881	0.881	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.881	0.881	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	LEX $\rightarrow$ LEX+0D	0.594	0.678	+0.085	[+0.067, +0.102]	no	0.00
TrQA Qw7B	LEX $\rightarrow$ LEX+SINK	0.594	0.728	+0.134	[+0.113, +0.154]	no	0.00
TrQA Qw7B	LEX $\rightarrow$ LEX+DEFL	0.594	0.678	+0.085	[+0.066, +0.103]	no	0.00
TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.847	0.842	-0.005	[-0.010, +0.001]	yes	1.00
TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.854	0.853	-0.001	[-0.001, -0.000]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.881	0.882	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.853	0.854	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.847	0.846	-0.000	[-0.003, +0.002]	yes	1.00
TrQA Qw7B	SINK $\rightarrow$ SINK(+0D)	0.713	0.732	+0.019	[+0.009, +0.029]	no	0.60

F INTERVENING ON THE SINK

Proposition 1 predicts that TOHA should track the response sink score more closely as the sink strengthens. We test this by intervening on the sink directly. For Qwen2.5-1.5B and TinyLlama on TruthfulQA, we add a constant $b \in \{-2, 0, 2, 4\}$ to the first token’s attention logit in every head and recompute all per-head quantities (Table 13). The intervention leaves the identity intact (no violations in any run). It moves the structure the proposition says matters. Suppressing the sink ($b=-2$) lowers the share of prompt-coned head graphs from 58% to 37% (Qwen) and from 74% to 63% (TinyLlama). It lowers the share where the sink is every response token’s top prompt token from 45% to 21% and from 63% to 21%, and the median rank correlation between d and the response sink score from 0.893 to 0.605 and from 0.967 to 0.406. Strengthening it ($b=4$) raises the correlation to 0.962 and 0.996. In TinyLlama all three quantities increase monotonically with b . In Qwen they peak at $b=2$ and fall slightly at $b=4$. TOHA’s own AUC barely moves (0.65 to 0.67 and 0.65 to 0.70). Its gap to the sink-score version is not monotone in b , so the intervention supports the mechanism at the level of the attention graph, not a dose-response in detection accuracy.

G MISTRAL-7B

Mistral-7B-Instruct-v0.2 is the most sink-dominated model in this study, and the reduction is correspondingly tightest there. Under its [INST] format it places 0.75 of its attention on token 0 — more than any smaller model — and 98.2% of its 52,288 layer graphs are *exactly* coned ($\delta_0 = 0$), with 0 violations of any bound in Theorem 1.

An earlier accuracy-level audit of this model reported that 99.99% of its 1D features are exactly zero and that the 0D and 0D+1D banks are equivalent, and inferred from Theorem 1(c) that near-universal coning would account for it — without checking δ_0 directly. Both halves now hold on the graphs themselves: 1D persistence is exactly zero on 99.99% of them, and $\delta_0 = 0$ is verified on 98.2%. The length-normalized rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ has median 1.000 over layers and never falls below 0.994, so on this model 0D total persistence is, up to rank, an affine function of sink attention in every layer. Per head, 81% of its graphs are exactly prompt-coned, the median $\rho(d, \bar{\pi})$ is 0.998, and Proposition 1 has 0 violations.

H RELATION TO ACCURACY-LEVEL AUDITS, AND COST

Audits of attention-graph PH that stop at classification accuracy observe four facts that the theory now explains: (i) 0D persistence equals an MST statistic (Fact 1); (ii) 1D persistence adds little beyond 0D (Table 10, column “1D | 0D”), which Theorem 1 now explains; (iii) an earlier audit reported that the 1D bank of Mistral-7B under its [INST] format is 99.99% zeros, which is what exact coning would produce (not yet re-verified with this pipeline; Appendix G); (iv) HaluEval answers are separable from text alone.

Cost. Median per-matrix wall-clock time on real attention from TinyLlama-1.1B-Chat-v1.0 (sink features / δ_0 / dense MST / ripser 0D / ripser 0D+1D) is 0.001/0.02/0.10/0.02/0.02 ms at $N=32$; 0.001/0.03/0.21/0.03/0.03 ms at $N=64$; 0.001/0.07/0.42/0.07/0.06 ms at $N=128$;

Table 10: **Key paired comparisons** (ΔAUC). “X–Y” is $\text{AUC}(\text{X}) - \text{AUC}(\text{Y})$; “X|Y” is $\text{AUC}(\text{Y}+\text{X}) - \text{AUC}(\text{Y})$ under early fusion; “X $\oplus$ Y” is the same under late fusion (stacked out-of-fold logits). $P_{0,h}$: per-head MST total. Defl.: sink-deflated 0D+1D PH; NT: all non-topological features. $\equiv$: equivalent within ± 0.015 (90% interval inside the margin); *: 95% interval excludes 0. Intervals and TOST power for every cell are in Appendix E.

Setting	Sink–0D	0D Sink	$P_{0,h}$ Sink _h	Defl. Sink	1D 0D	0D NT	0D $\oplus$ NT	Defl. NT
TQA Qw3B	+0.001 $\equiv$	+0.003 $\equiv$	+0.009 $\equiv$	+0.029*	+0.033*	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw1.5B	–0.003 $\equiv$	+0.012*	+0.011 $\equiv$	+0.034*	–0.008 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Phi-3	+0.005 $\equiv$	+0.006 $\equiv$	0.000 $\equiv$	+0.033*	–0.003 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA TinyLL	–0.010	+0.020*	+0.003 $\equiv$	+0.045*	+0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA SmoLLM	+0.028*	+0.030*	+0.106*	+0.041*	+0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA Mis7B	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.055*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Mis7B (chat)	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.067*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Qw7B	+0.070*	+0.027*	+0.060*	+0.038*	+0.021*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw3B	+0.018*	+0.028*	+0.004 $\equiv$	+0.058*	+0.015*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	+0.031*	+0.012*	+0.008 $\equiv$	+0.052*	+0.026*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	–0.015*	+0.021*	+0.004 $\equiv$	+0.002	–0.003 $\equiv$	+0.001 $\equiv$	+0.002 $\equiv$	0.000 $\equiv$
TrQA Qw1.5B	–0.009	+0.010	+0.004 $\equiv$	+0.012	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TrQA Phi-3	–0.004 $\equiv$	+0.006 $\equiv$	+0.002 $\equiv$	+0.029*	–0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.002 $\equiv$
TrQA TinyLL	+0.001 $\equiv$	+0.004 $\equiv$	–0.001 $\equiv$	+0.014	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Mis7B	–0.003 $\equiv$	+0.002 $\equiv$	+0.002 $\equiv$	+0.043*	–0.007	0.000 $\equiv$	+0.003 $\equiv$	0.000 $\equiv$
TrQA Qw7B	+0.046*	+0.022*	+0.031*	+0.020*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$

Table 13: **Sink-bias intervention** (teacher-forced TruthfulQA; labels unchanged). A constant b is added to the attention logit of key 0 for every query after token 0, in every head and layer. $\delta_P=0$: share of prompt-coned head graphs. Sink top: share of head graphs in which token 0 is every response token’s most-attended prompt token. ρ : median per-head rank correlation between TOHA’s d and the response sink score. Remaining columns: AUCs as in Table 3.

Model	b	$\delta_P=0$	sink top	$\rho(d, 1 - \bar{a}_0)$	TOHA	TOHA[$\bar{a}_0$]	TOHA _{sup}	$\bar{a}_{0,\text{sup}}$
Qwen2.5-1.5B	–2	37%	21%	0.605	0.667	0.624	0.843	0.849
Qwen2.5-1.5B	0	58%	45%	0.893	0.646	0.628	0.858	0.857
Qwen2.5-1.5B	+2	72%	60%	0.970	0.667	0.637	0.857	0.857
Qwen2.5-1.5B	+4	69%	56%	0.962	0.647	0.640	0.839	0.838
TinyLlama-1.1B	–2	63%	21%	0.406	0.698	0.626	0.845	0.834
TinyLlama-1.1B	0	74%	63%	0.967	0.666	0.618	0.845	0.848
TinyLlama-1.1B	+2	81%	75%	0.992	0.652	0.610	0.848	0.848
TinyLlama-1.1B	+4	86%	83%	0.996	0.665	0.616	0.844	0.842

0.001/0.25/0.90/0.18/0.18 ms at $N=256$. On i.i.d. random matrices of the same size, ripser 0D+1D takes 0.171 ms ($N=32$), 3.46 ms ($N=64$), 115 ms ($N=128$), 4,631 ms ($N=256$). On real attention the same computation is four orders of magnitude cheaper at $N=256$, consistent with Theorem 1: sink-dominated graphs have few, short 1D bars to compute. Sink features, read off one column, are one to two orders of magnitude cheaper still.